Multimodal Machine Learning

Lecture 5.1: Multimodal Transformers (Part 1)

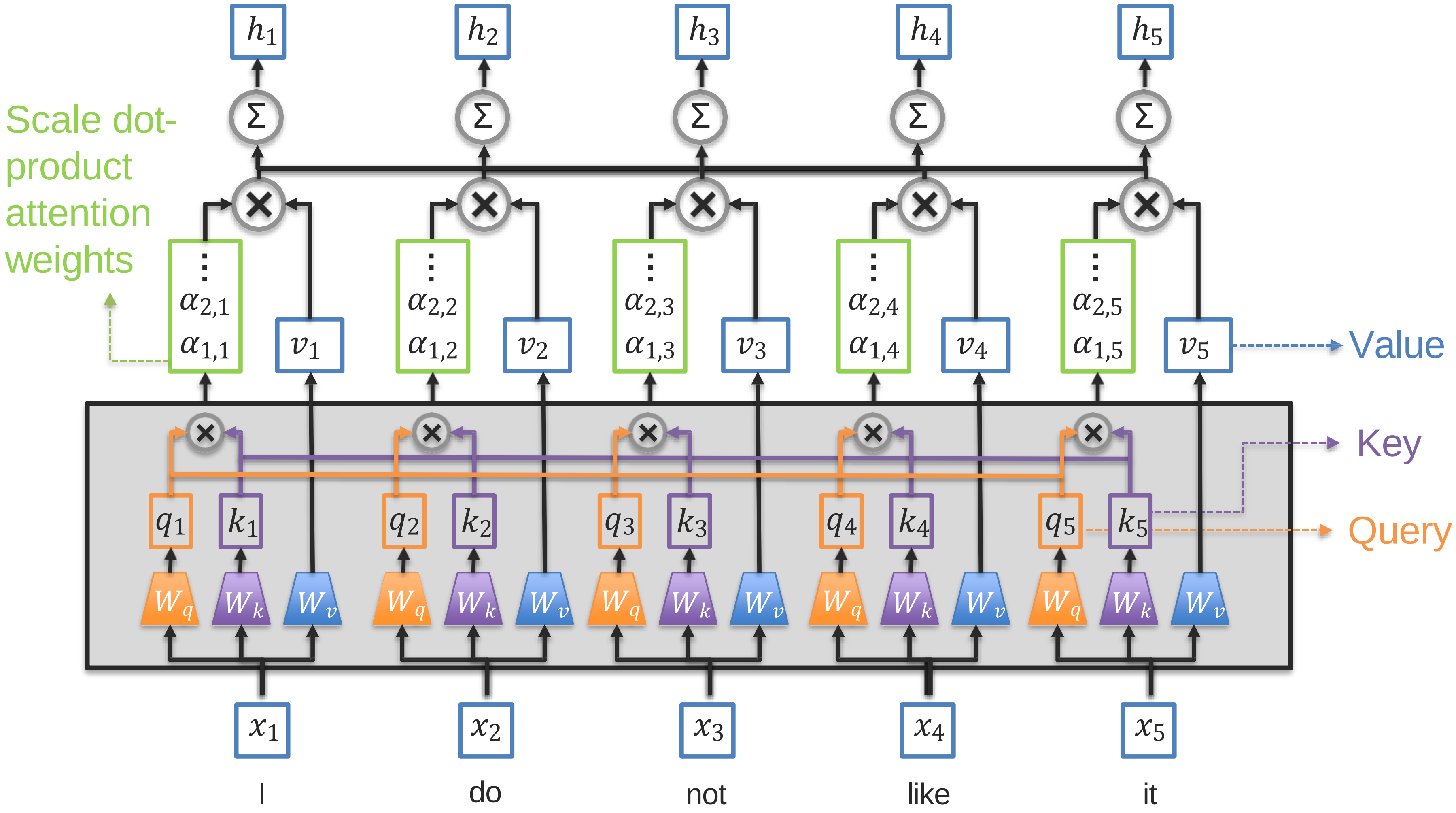
Dam Quang Tuan

Objectives of today's class

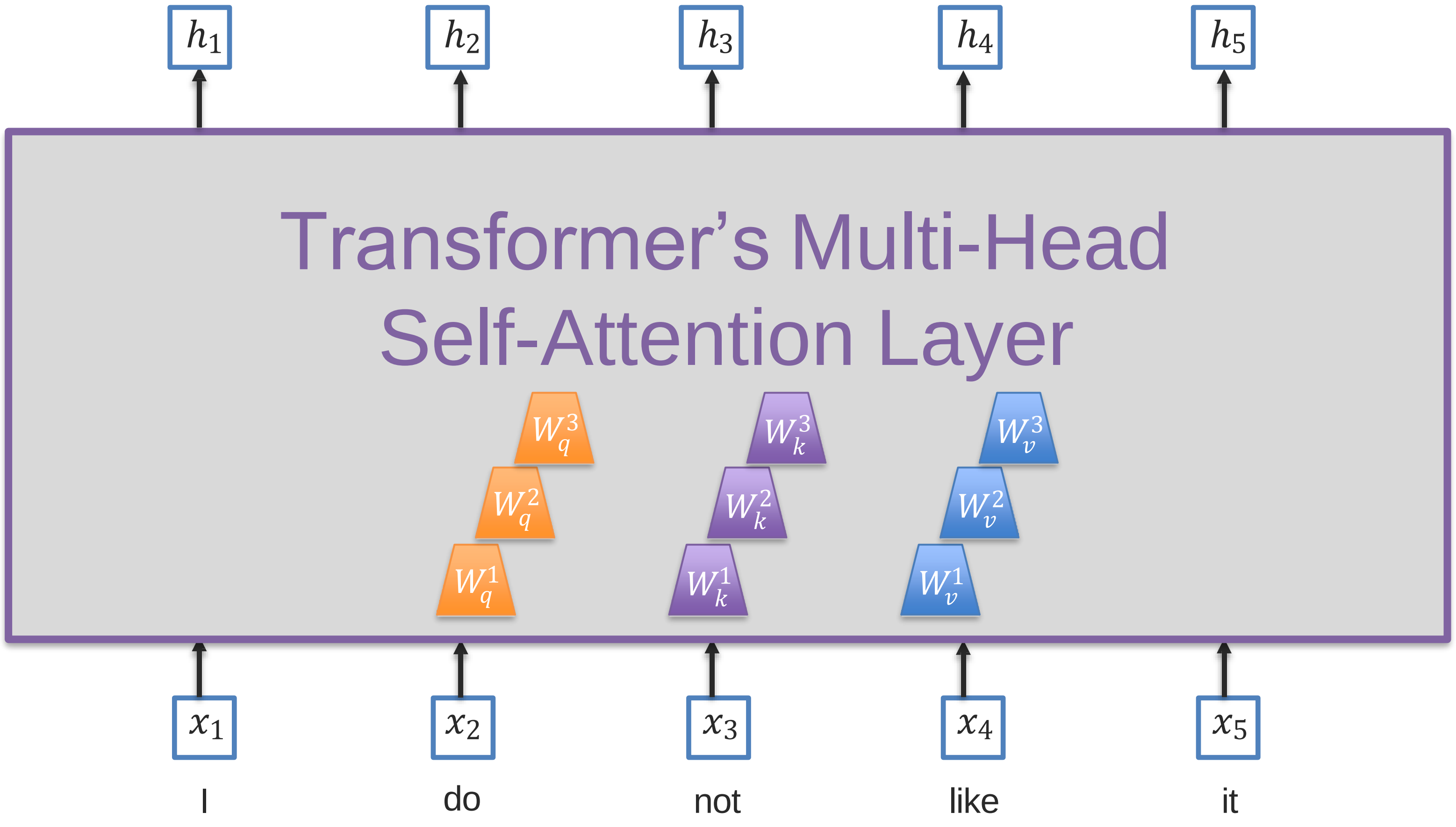
- Positional embeddings
- Language pre-training
  - BERT: Bidirectional Encoder Representations from Transformers
- Multimodal transformers (Image and language)
  - Concatenated transformers (VisualBERT, Uniter)
  - Crossmodal transformers (ViLBERT, LXMERT
  - Modality-shift transformer (MAG-BERT)
- Sequence-to-sequence modeling with Transformers

Positional Embeddings

Transformer Self-Attention



Transformer Multi-Head Self-Attention



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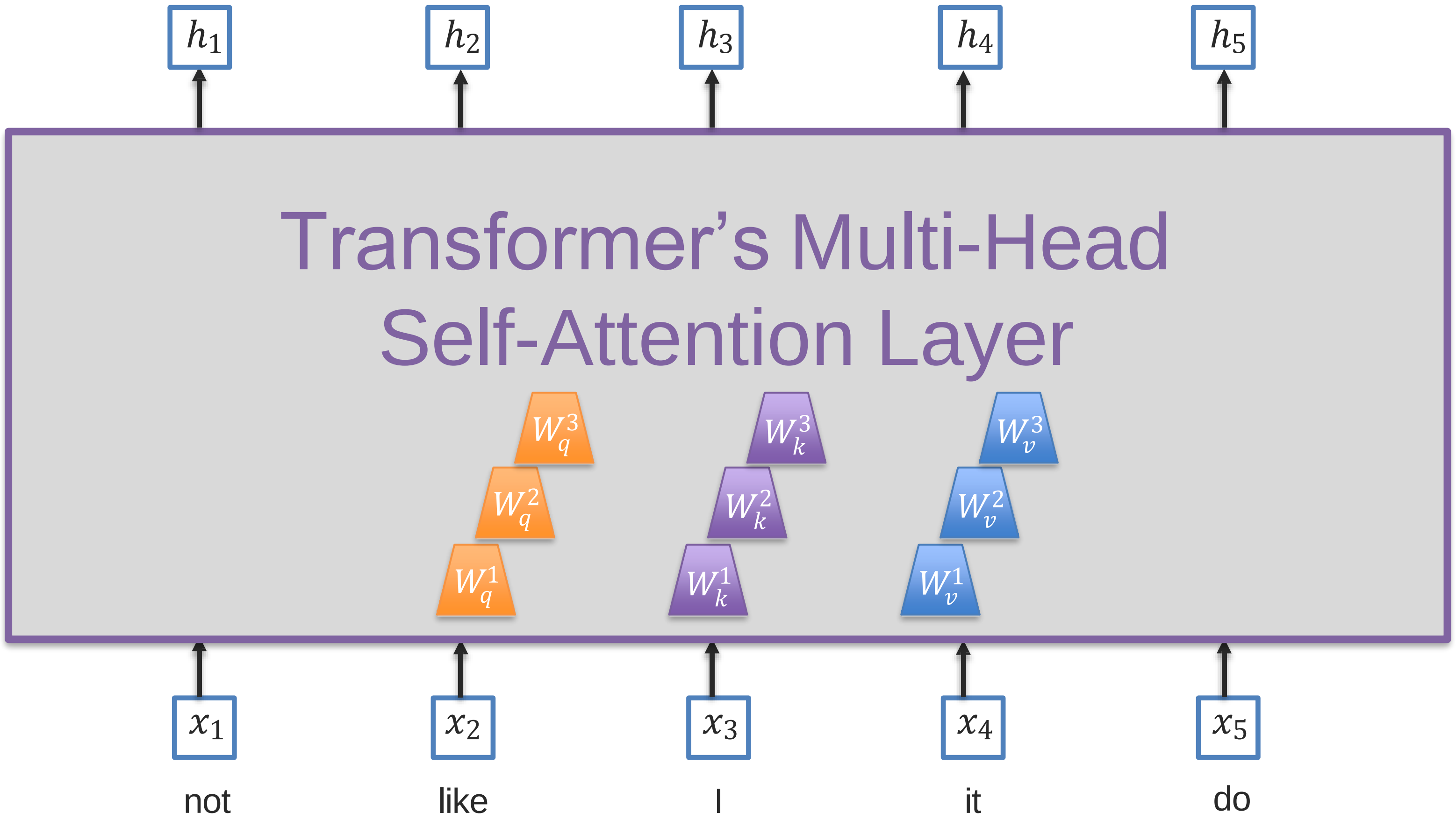


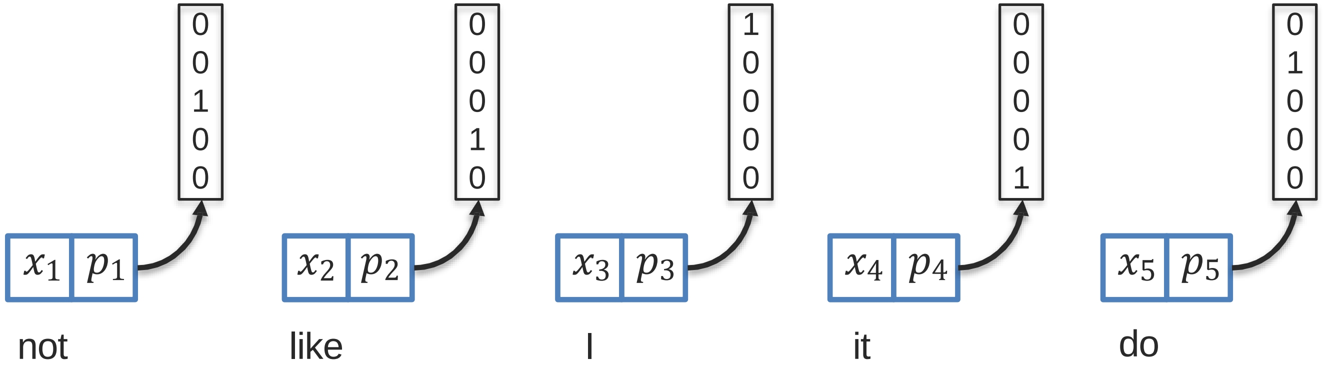
What happens if the words are shuffled?

Position embeddings

- Position information is not encoded in a self-attention module

How can we encode position information?

Simple approach: one-hot encoding

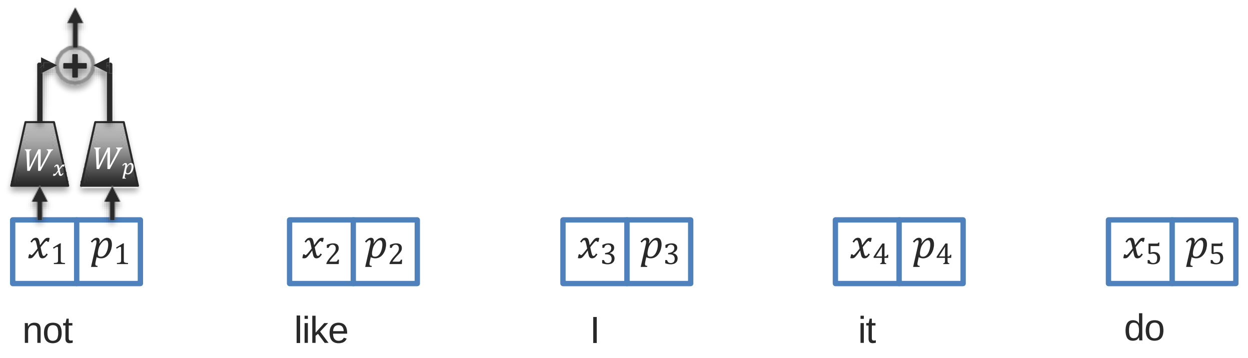


Position embeddings

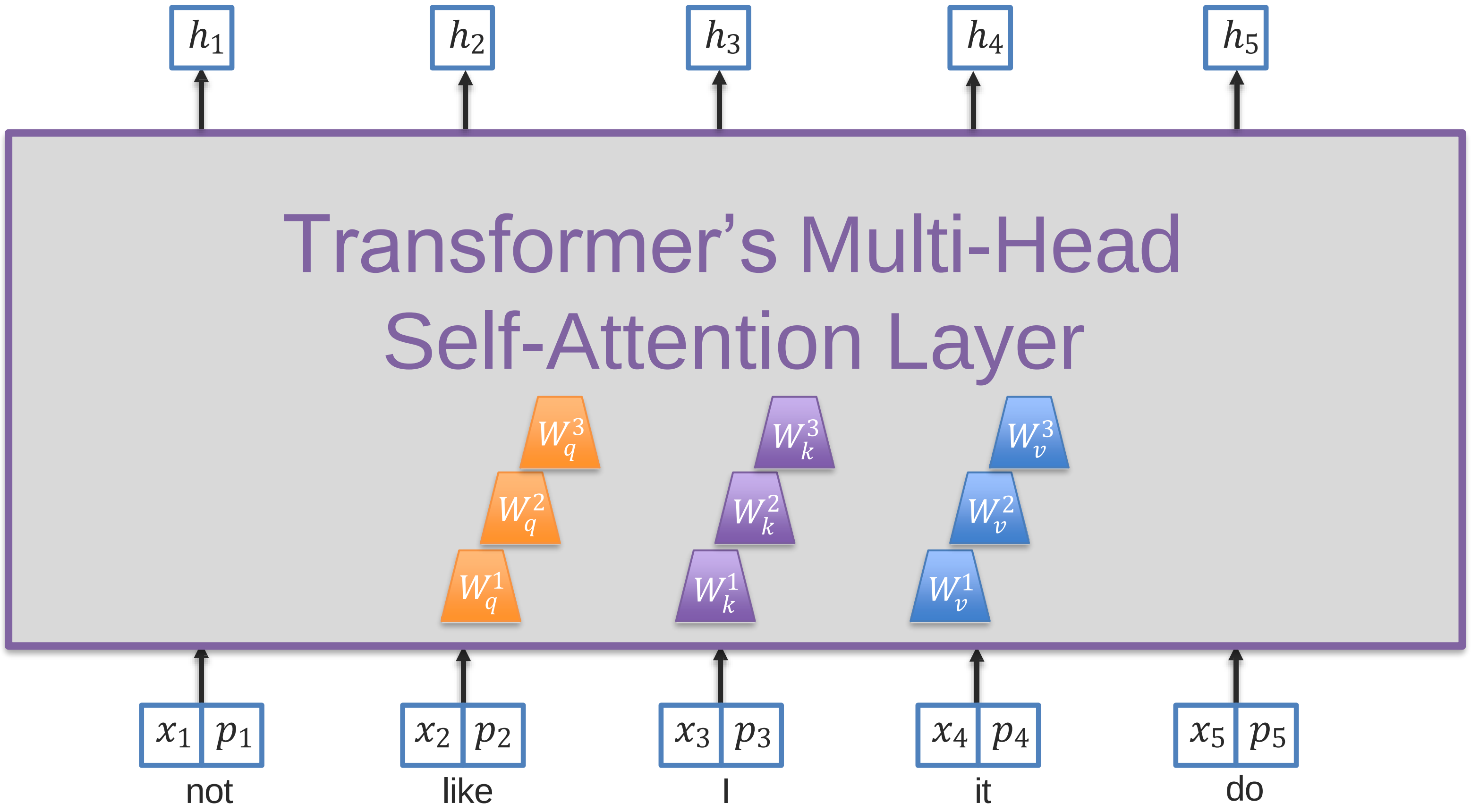
- ❑ Position information is not encoded in a self-attention module

How can we encode position information?

Simple approach: one-hot encoding + linear embeddings + $\left\{ \begin{array}{l} \text{Sum} \\ \text{- or -} \\ \text{concat} \end{array} \right.$

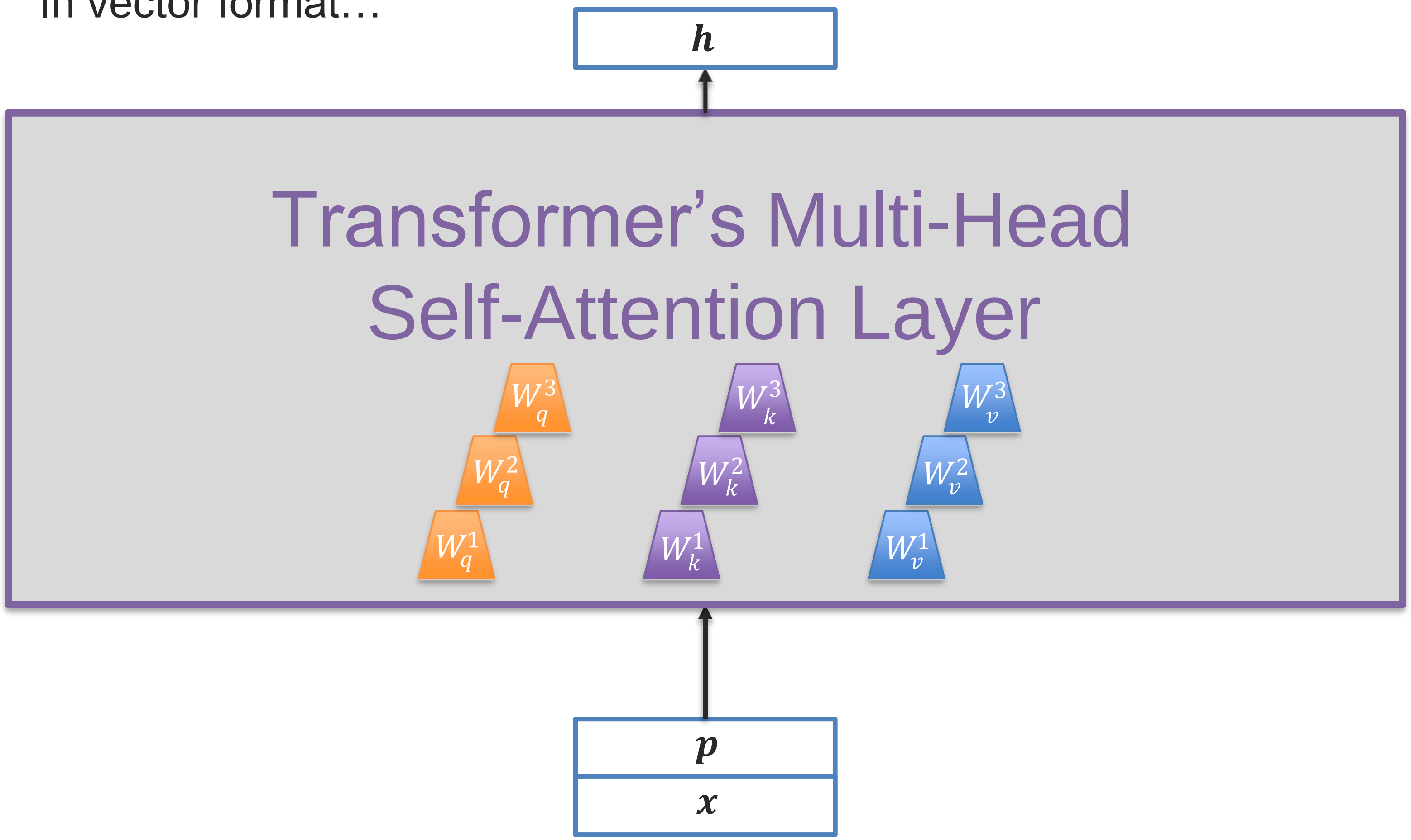


Transformer Multi-Head Self-Attention

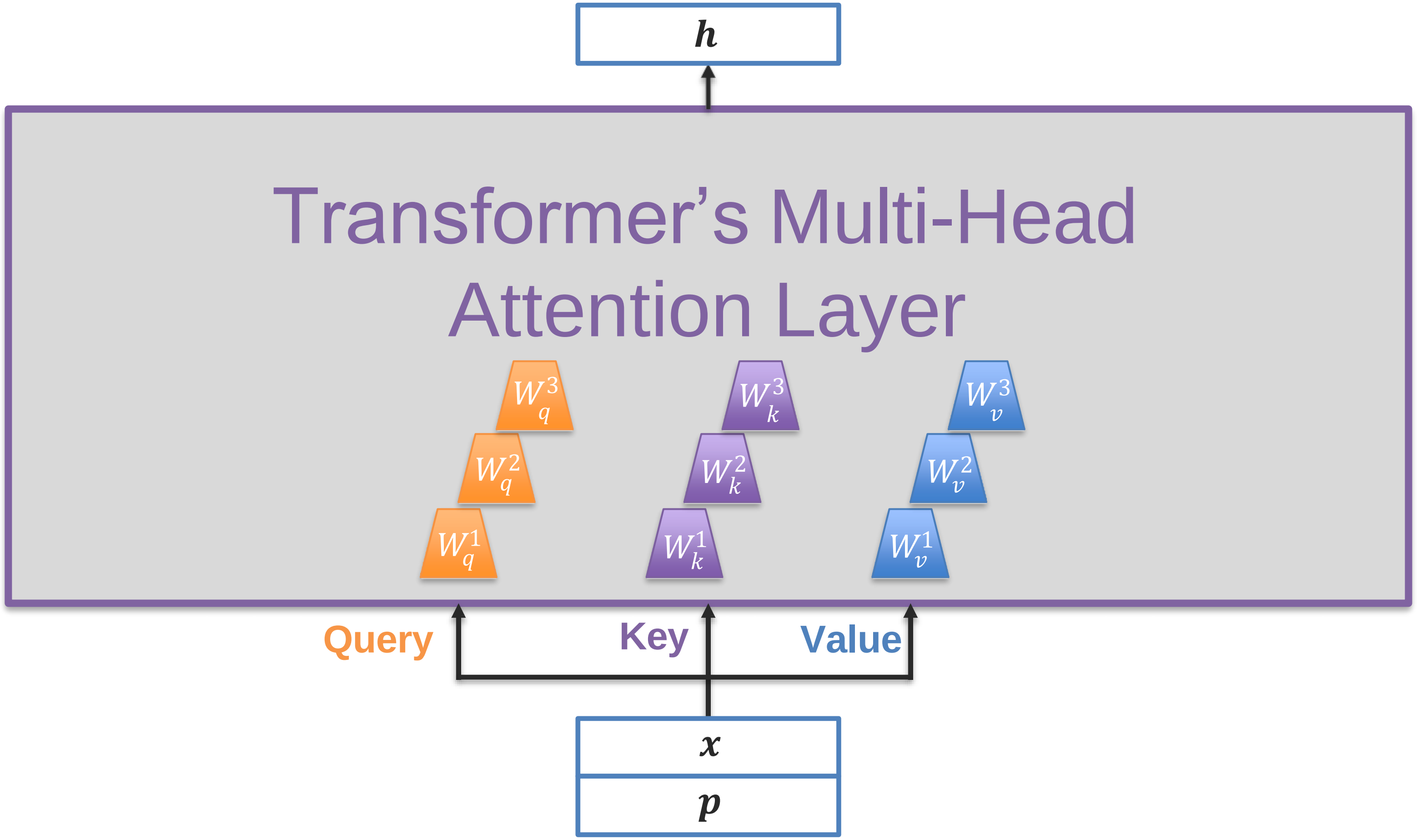


Transformer Multi-Head Self-Attention

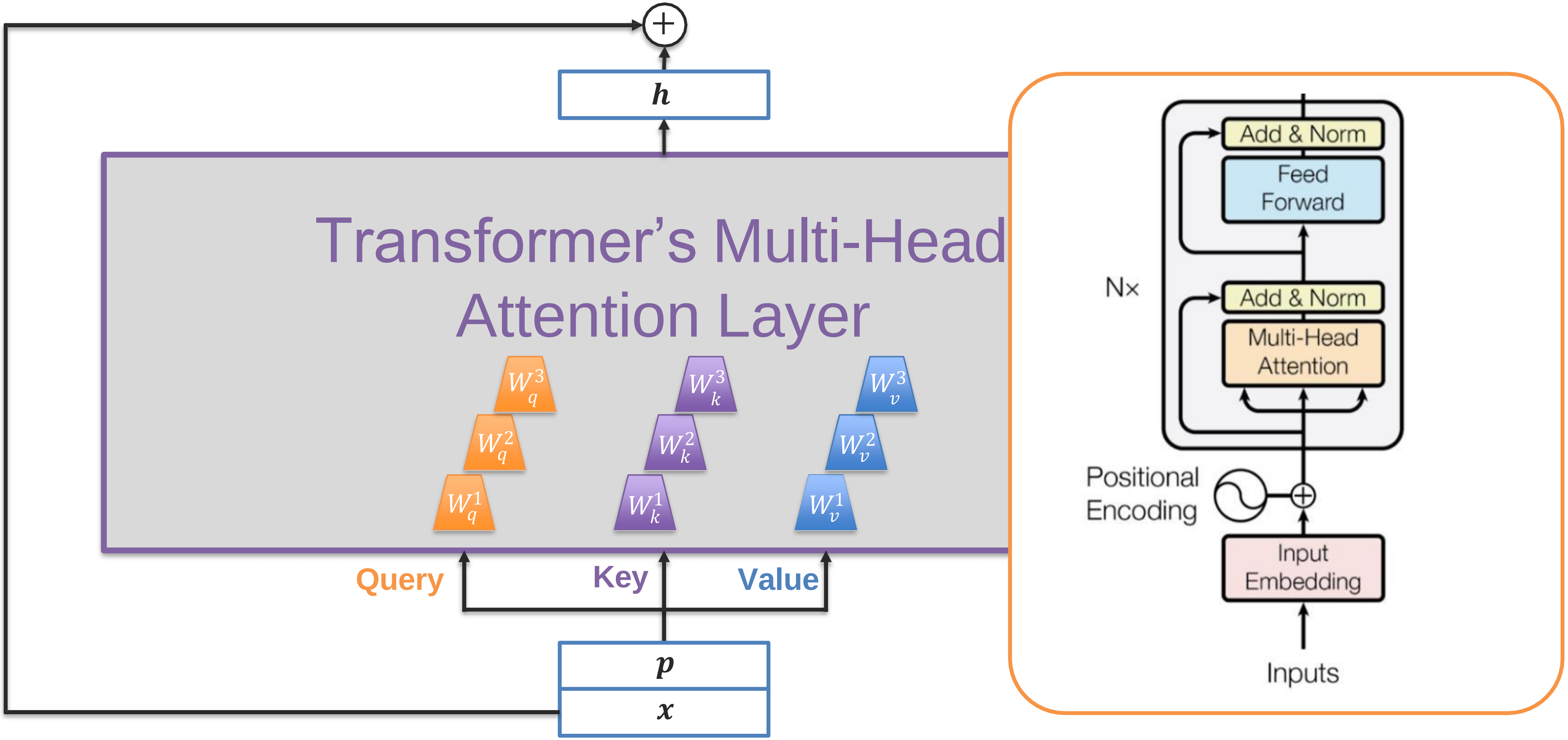
In vector format…



Transformer Multi-Head Attention



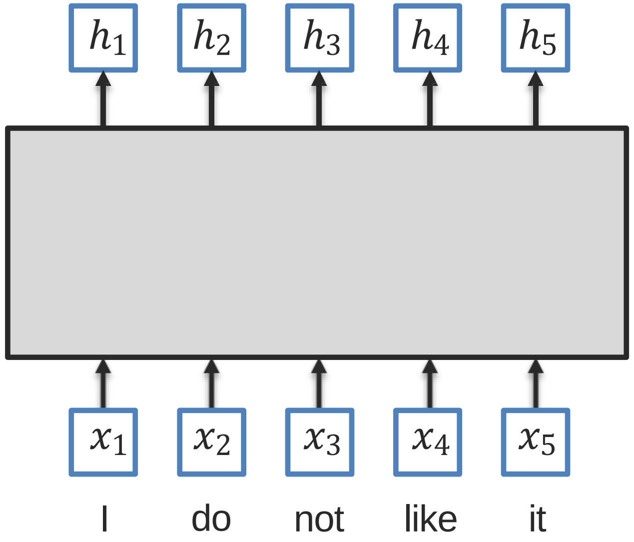
Transformer – Residual Connection



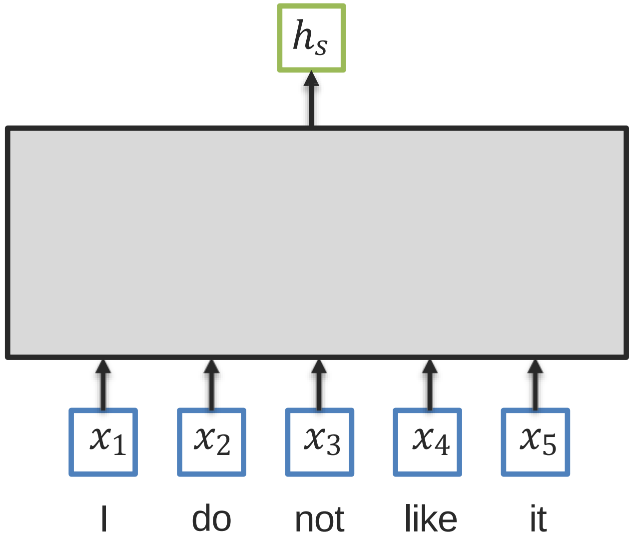
Language Pre-training

Token-level and Sentence-level Embeddings

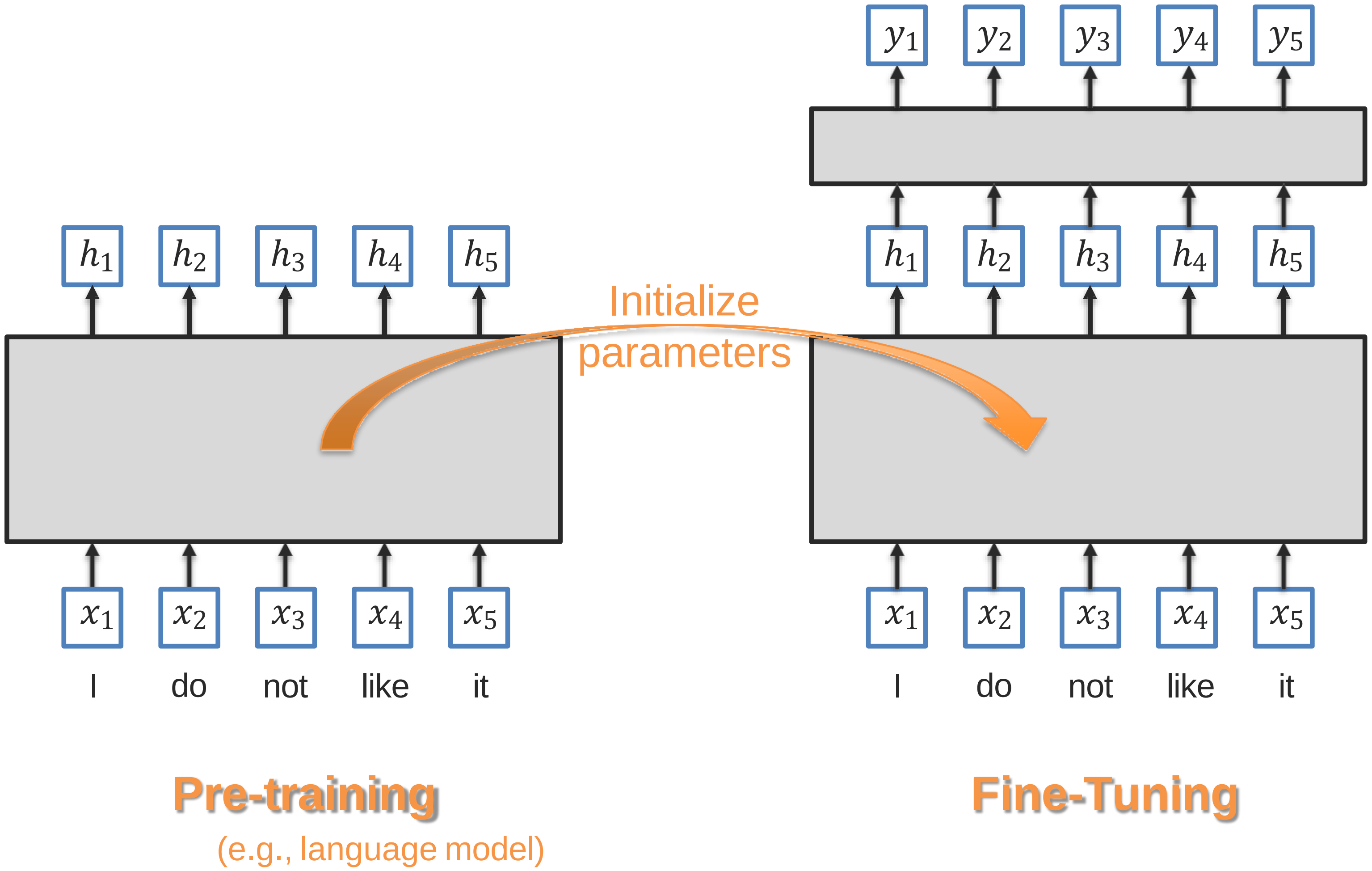
Token-level embeddings



Sentence-level embedding



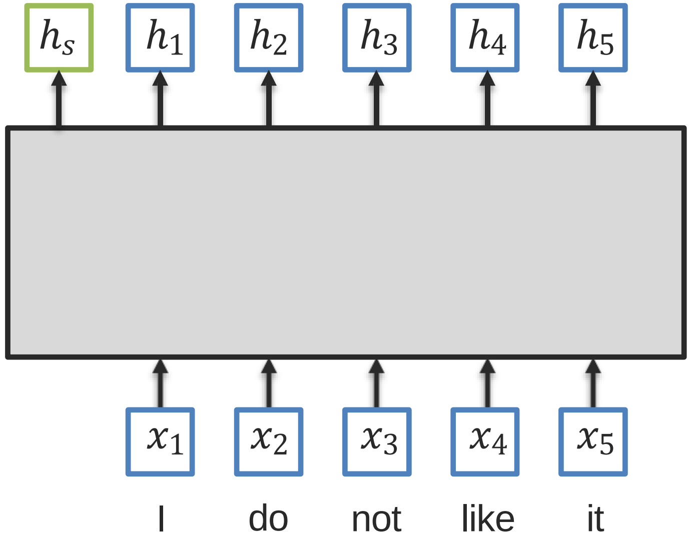
Pre-Training and Fine-Tuning



BERT: Bidirectional Encoder Representations from Transformers

Advantages:

1. Jointly learn representation for token-level and sentence level
2. Same network architecture for pre-training and fine-tuning



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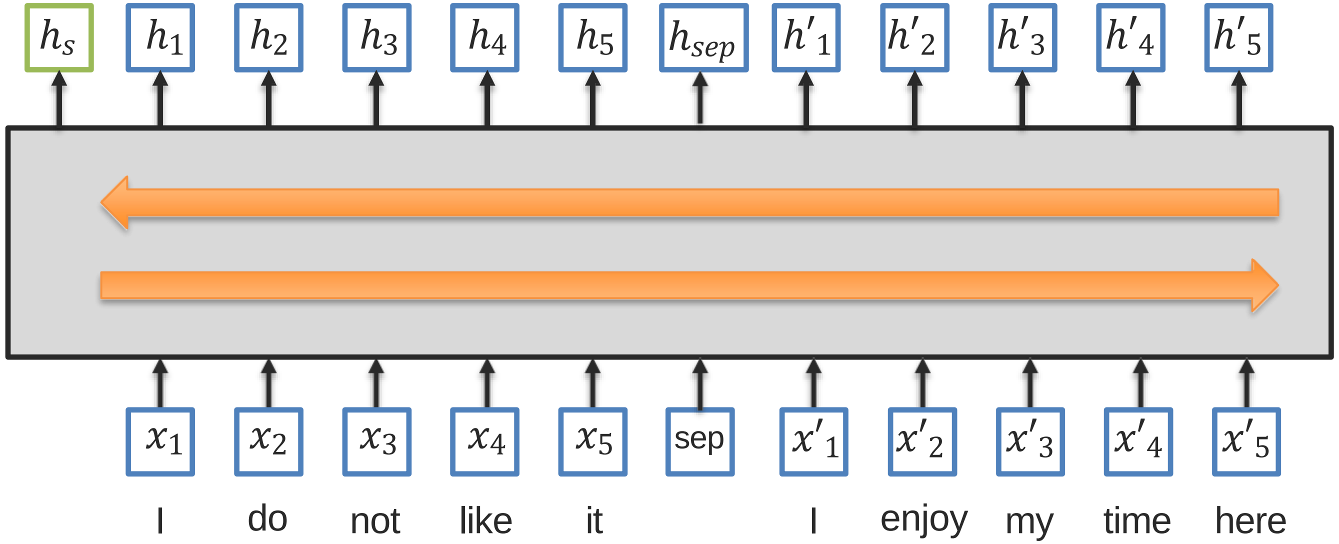
Advantages:

1. Jointly learn representation for token-level and sentence level
2. Same network architecture for pre-training and fine-tuning
3. Can be used learn relationship between sentences
4. Models bidirectional and long-range interactions between tokens

How can we do all this?



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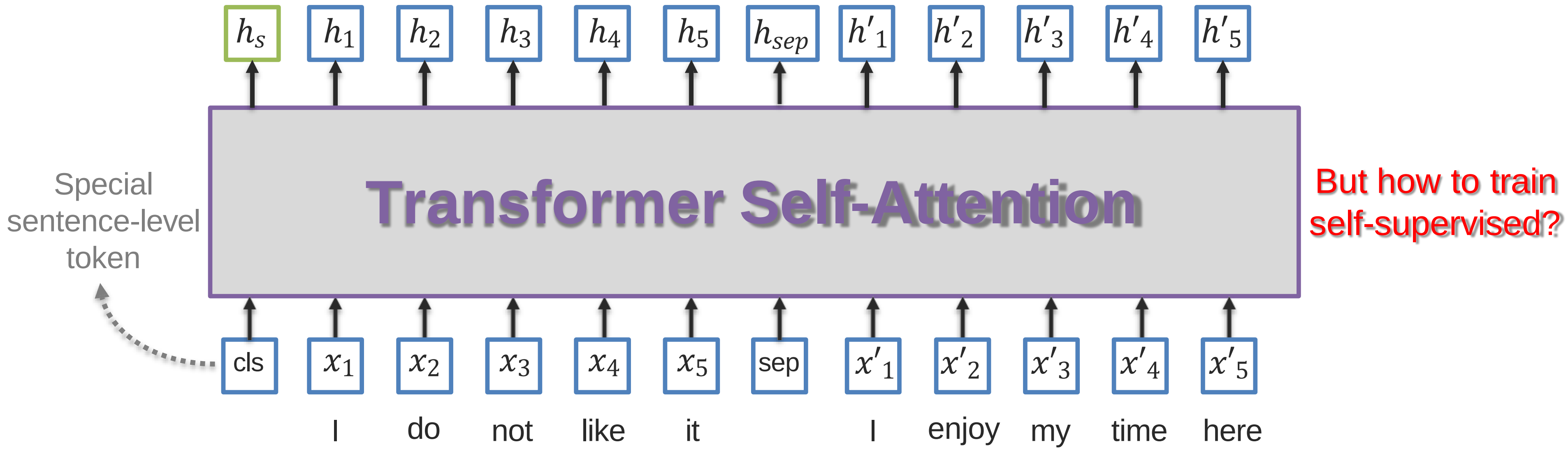
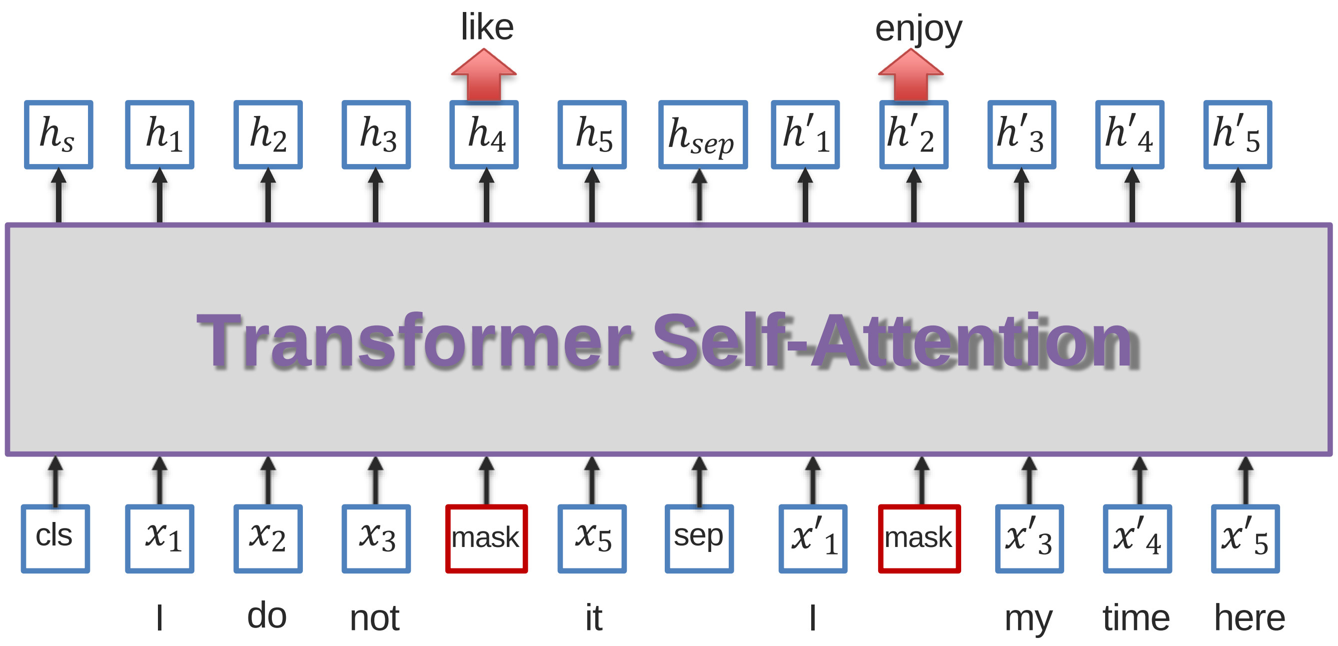


Pre-training BERT Model

1 **Masked Language Model**

Randomly mask input tokens and then try to predict them

What is the loss function?



Pre-training BERT Model

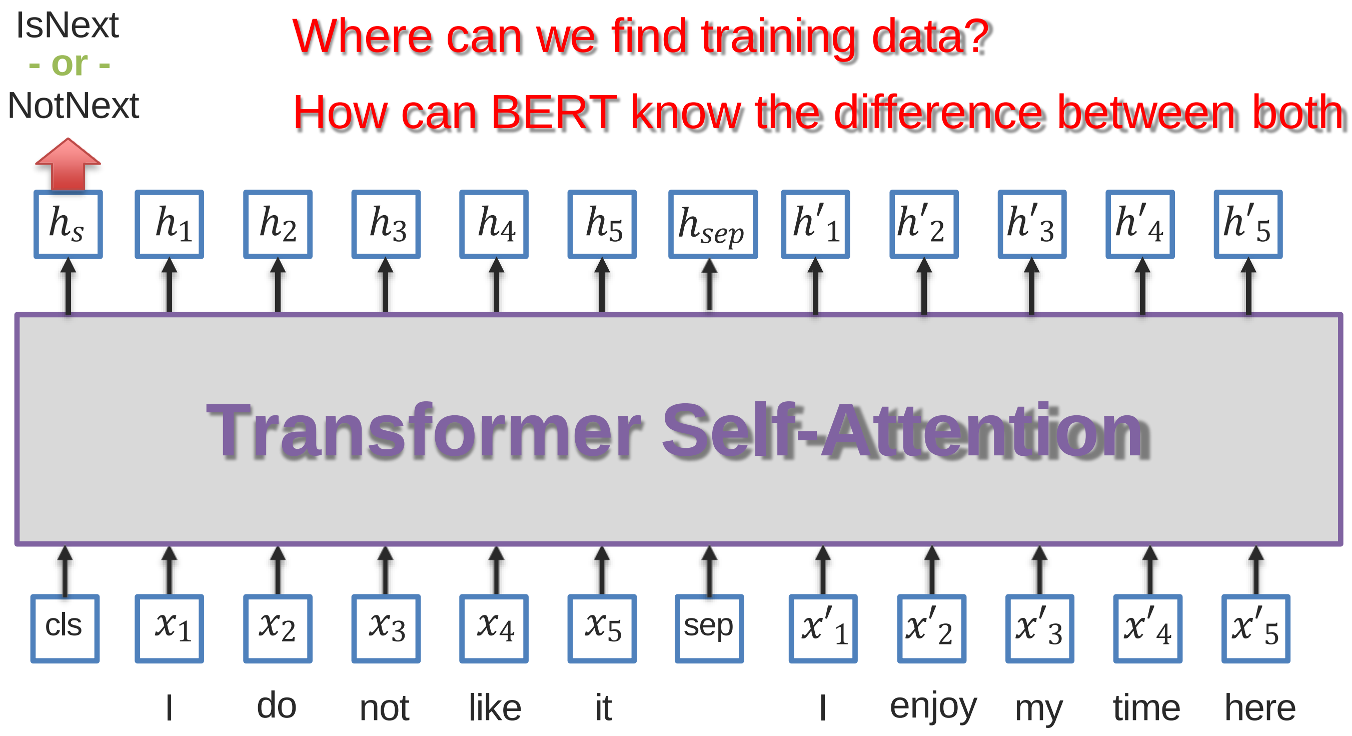
2 **Next Sentence Prediction**

Given two sentences, predict if this is the next one or not

What is the loss function?

Where can we find training data?

How can BERT know the difference between both sentences?



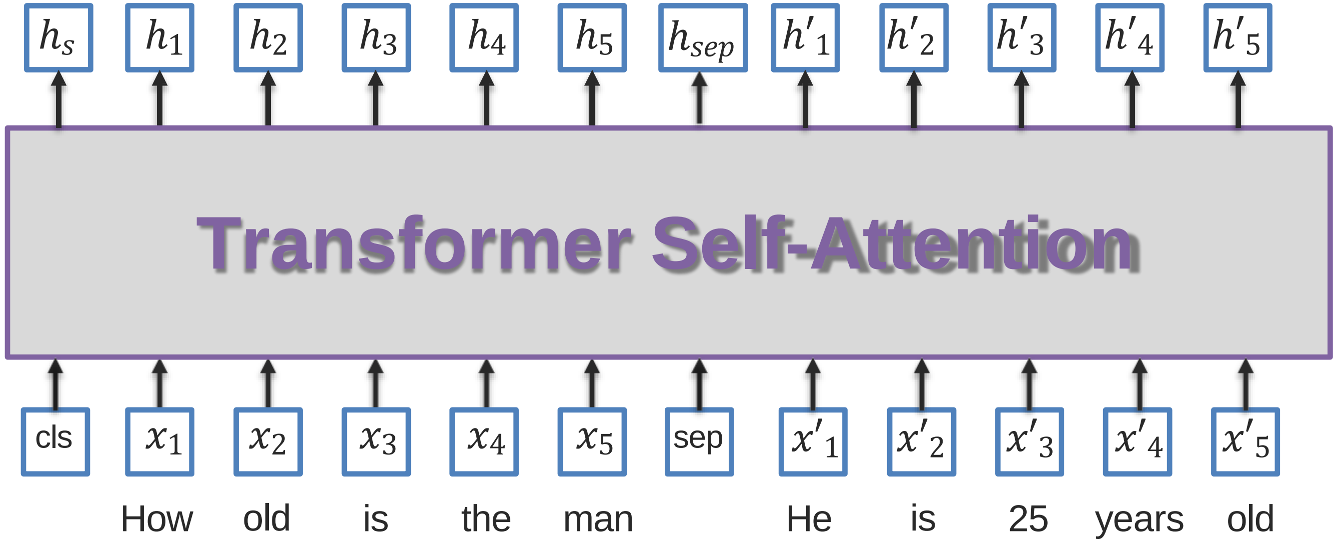
Fine-Tuning BERT

4 Question-answering: find start/end of the answer in the document

Paragraph: "*...Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.*"

Question 1: "*Which laws faced significant opposition?*"
Plausible Answer: *later laws*

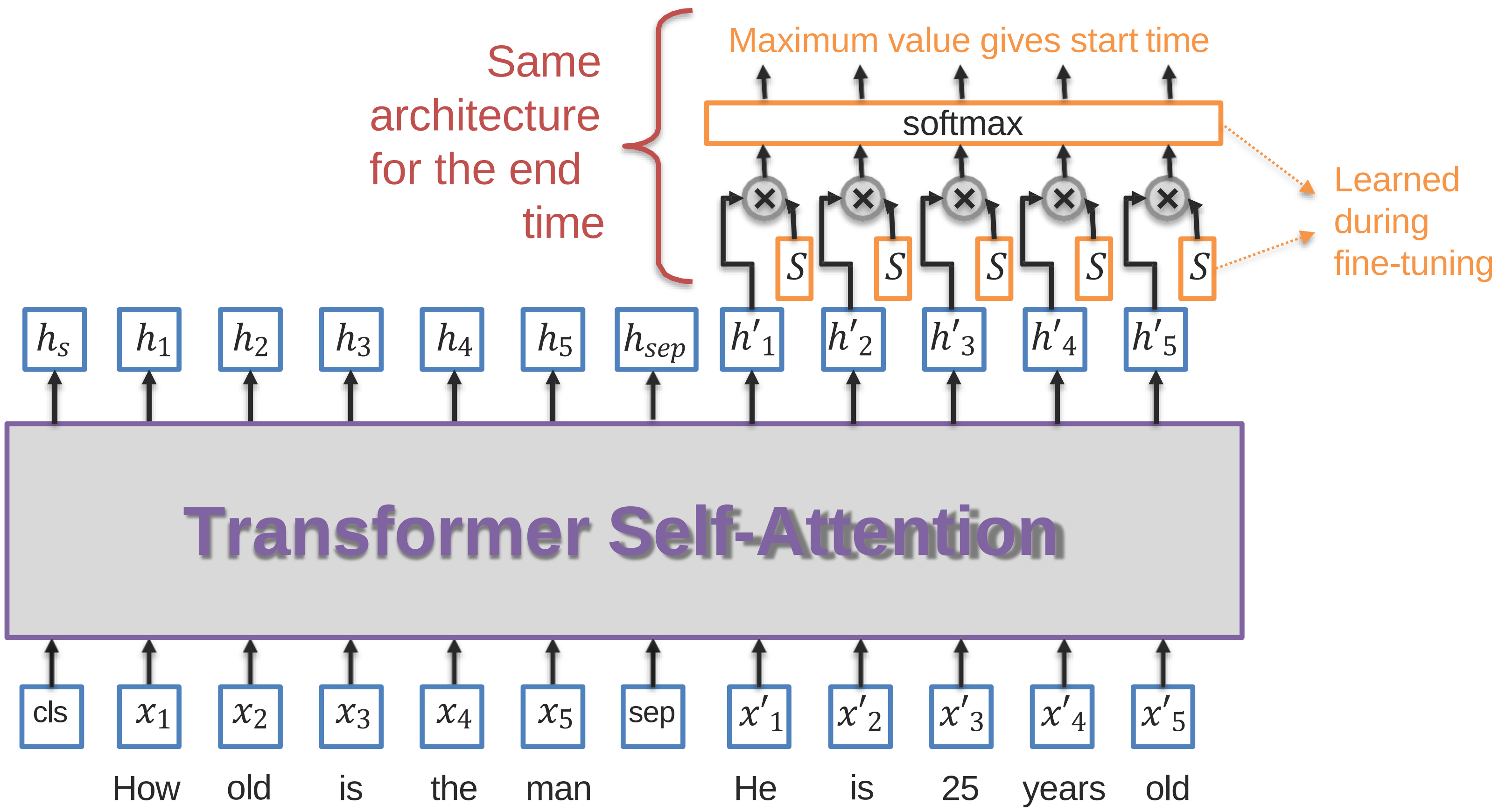
Question 2: "*What was the name of the 1937 treaty?*"
Plausible Answer: *Bald Eagle Protection Act*



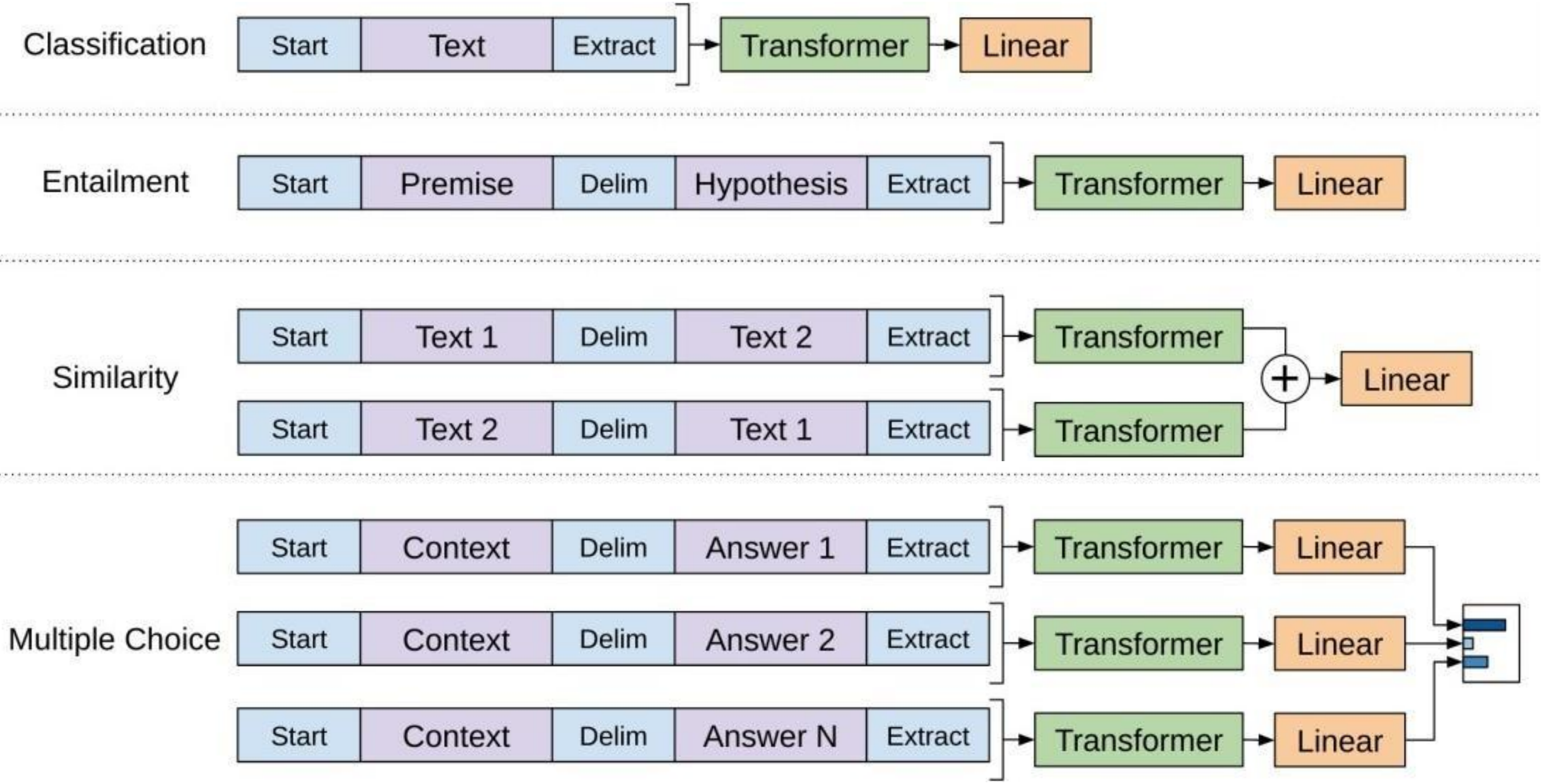
How?

Fine-Tuning BERT

4 Question-answering: find start/end of the answer in the document



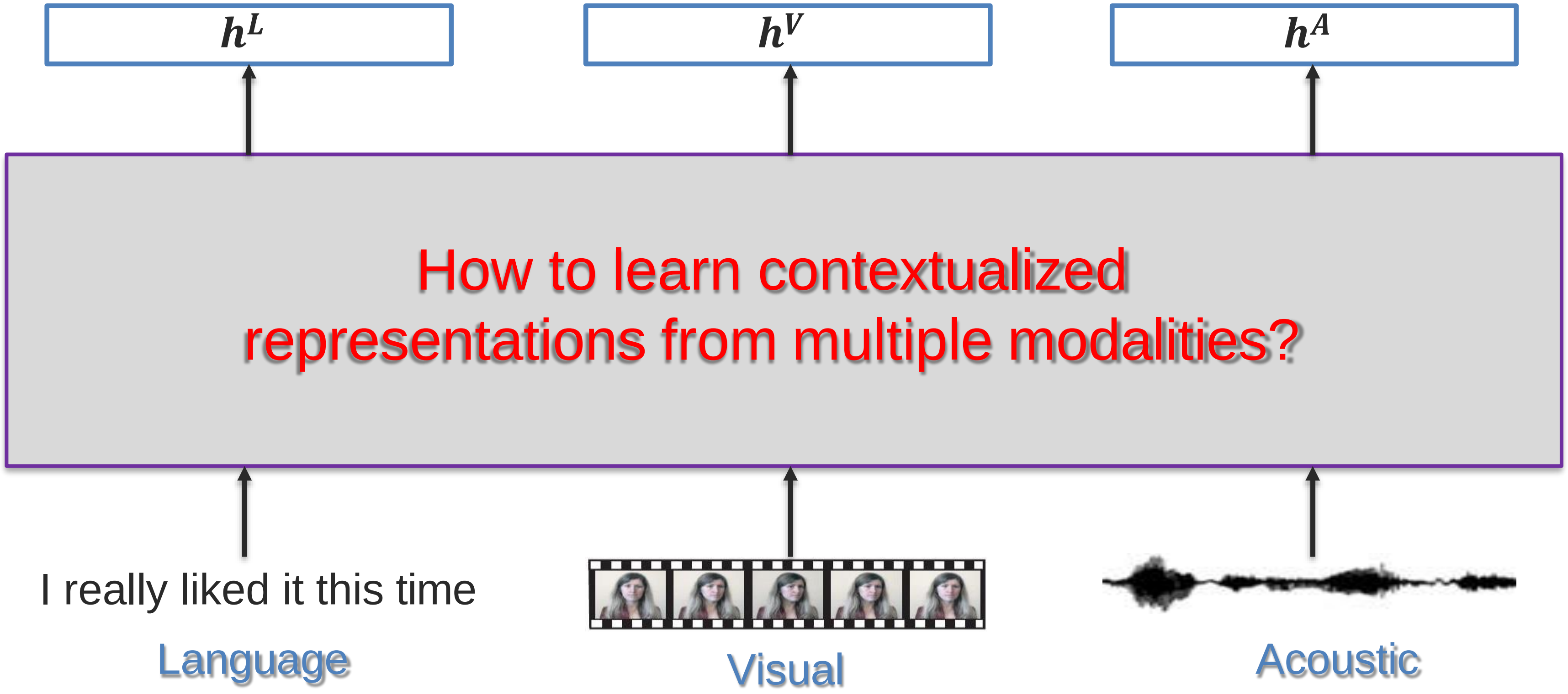
Other Fine-tuning Approaches



https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

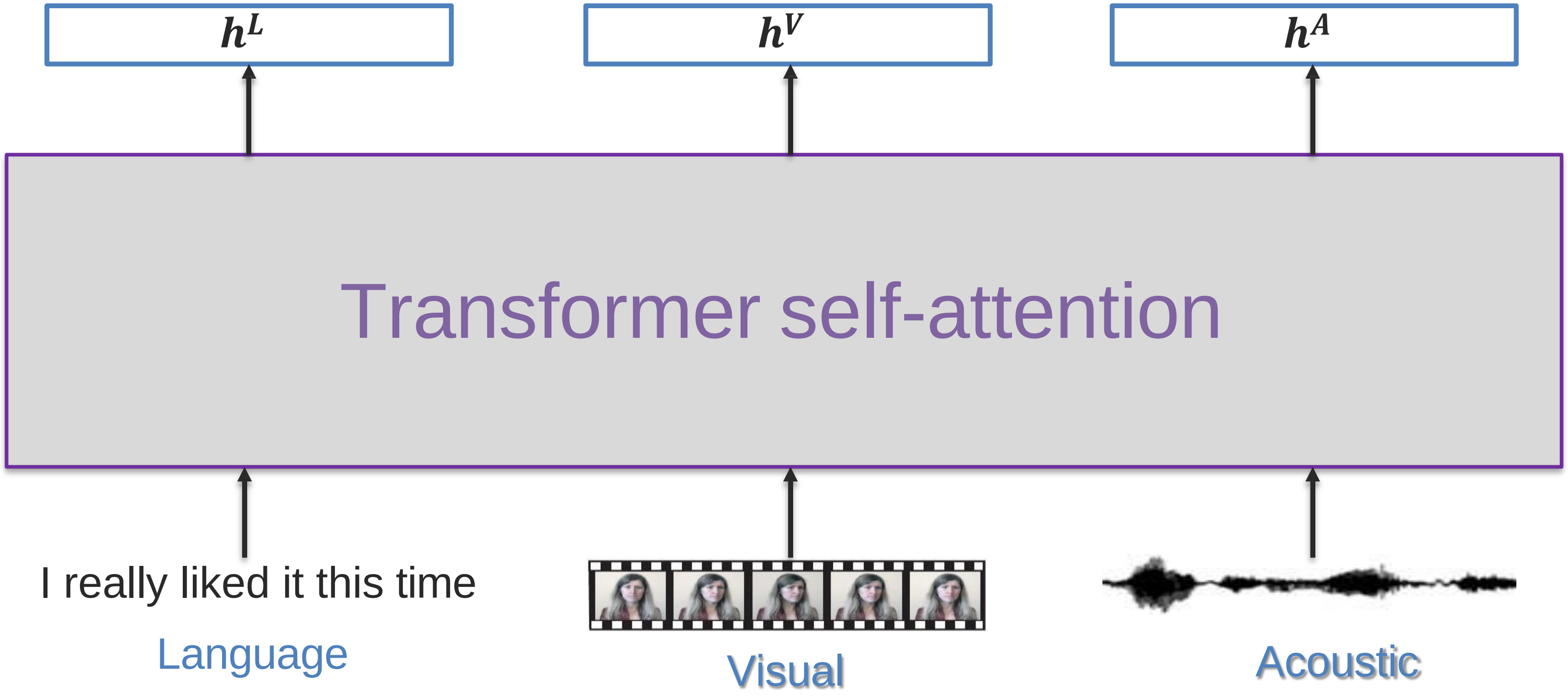
Language-Vision Transformers

Multimodal Embeddings

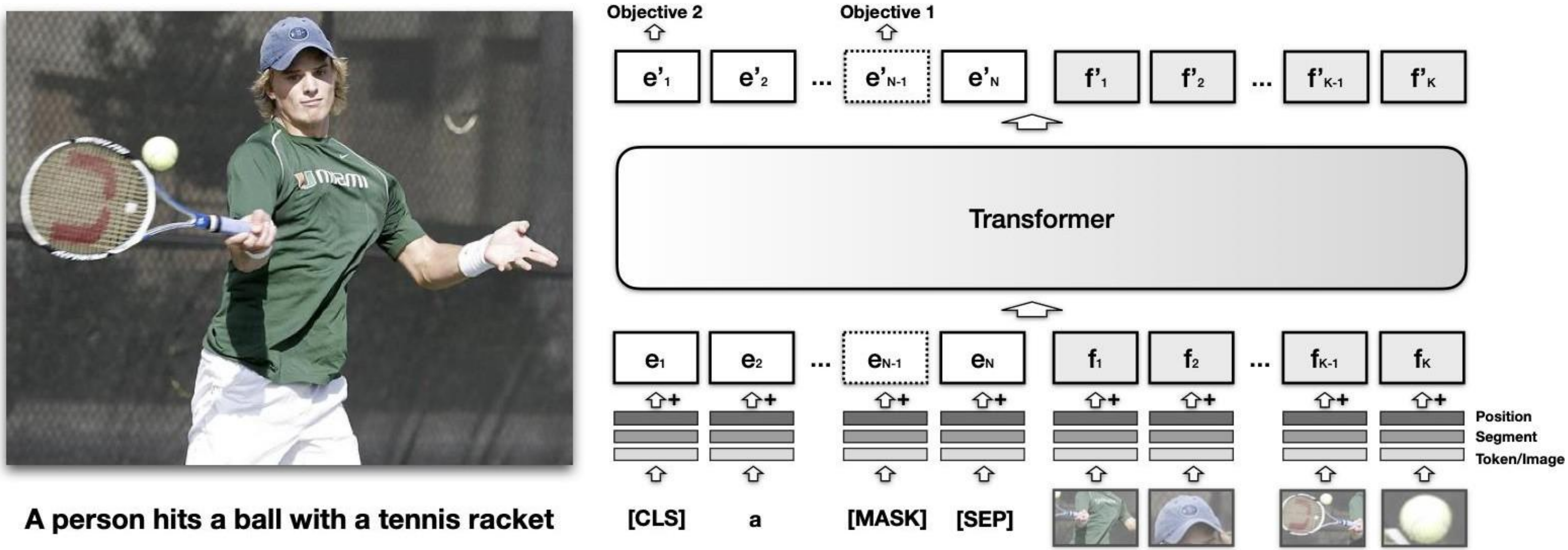


Option 1: Concatenate modalities and learn BERT transformer

Simple Solution: Contextualized Multimodal Embeddings



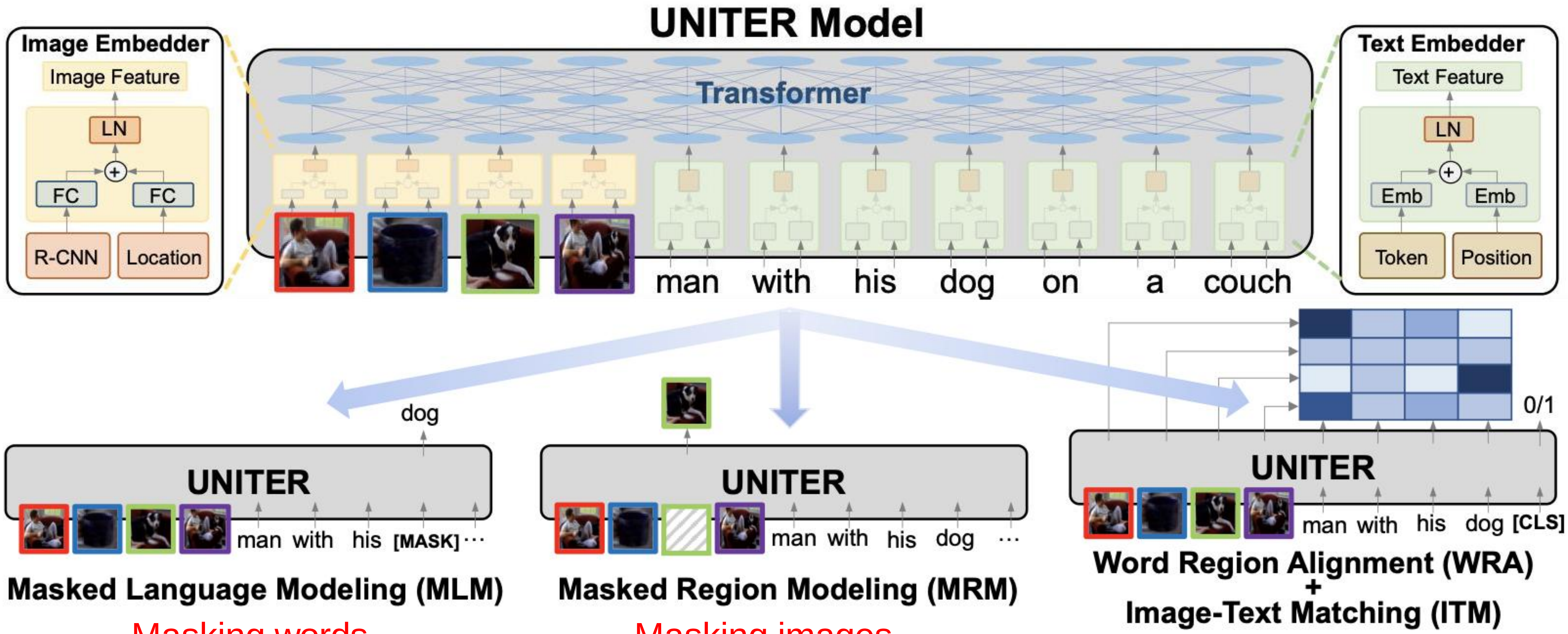
VisualBERT



Li, Liunian Harold, et al. "Visualbert: A simple and performant baseline for vision and language." *arXiv* (2019).

UNITER

Similar Transformer architecture to BERT and VisualBERT… but with slightly different optimization



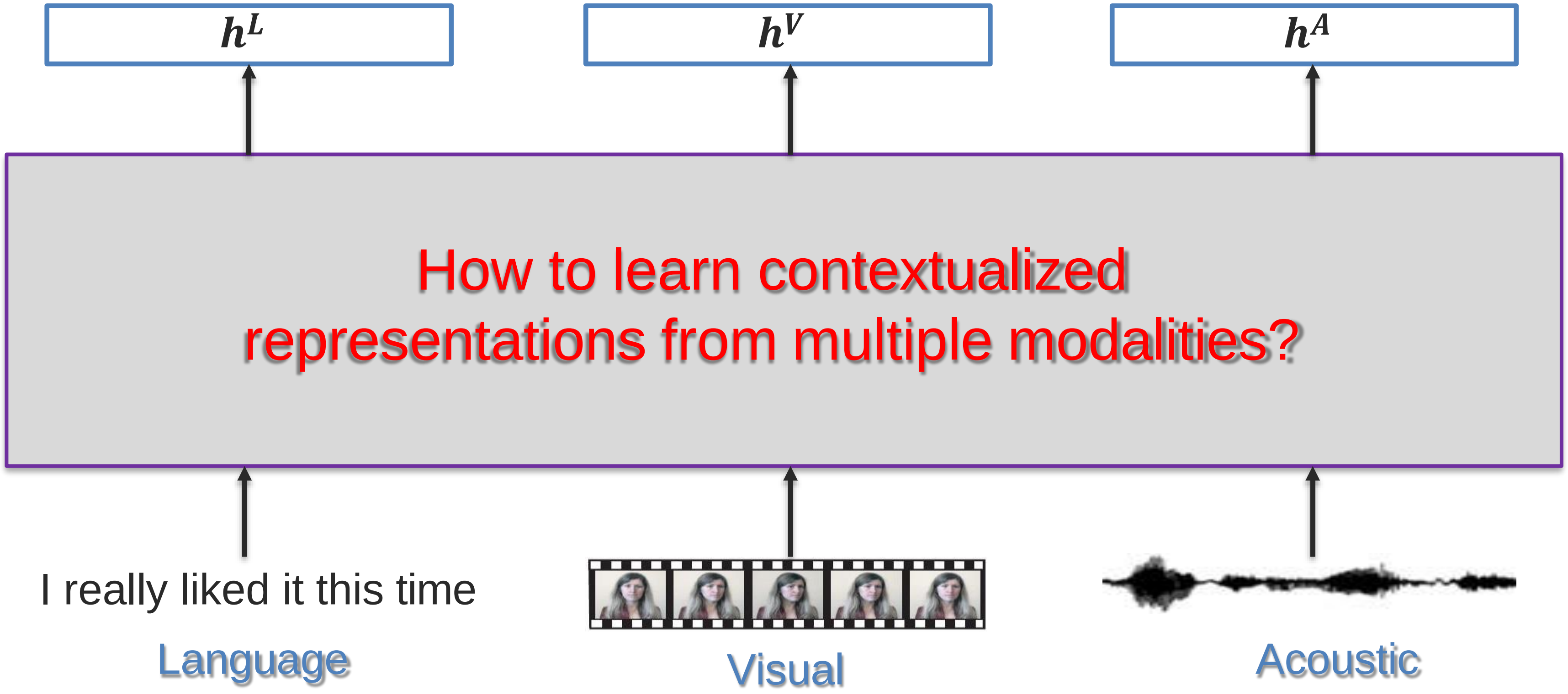
Masking words

Masking images

Is the sentence matches the image?

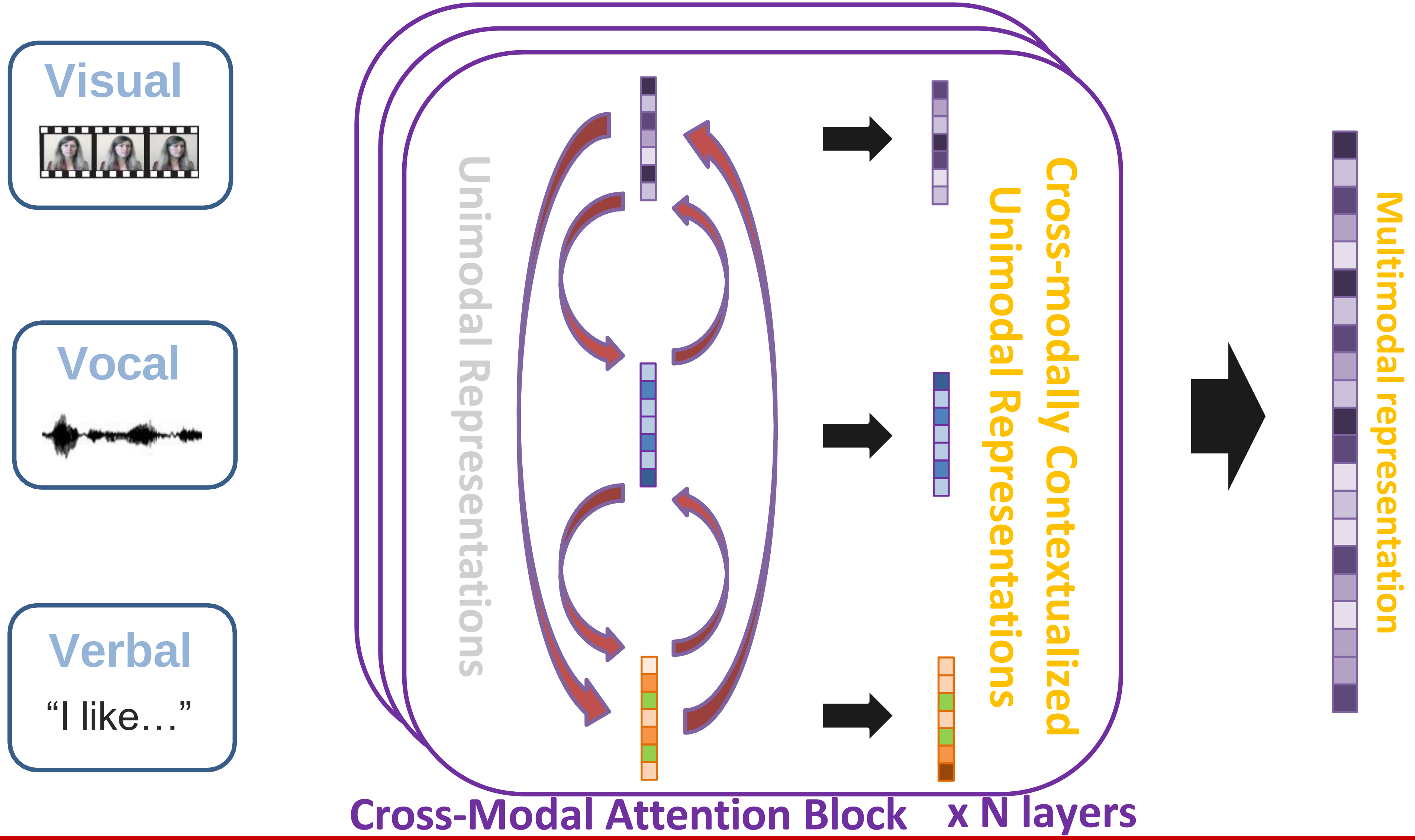
Chen, Yen-Chun, et al. "Uniter: Universal image-text representation learning." *European conference on computer vision*. 2020.

Multimodal Embeddings

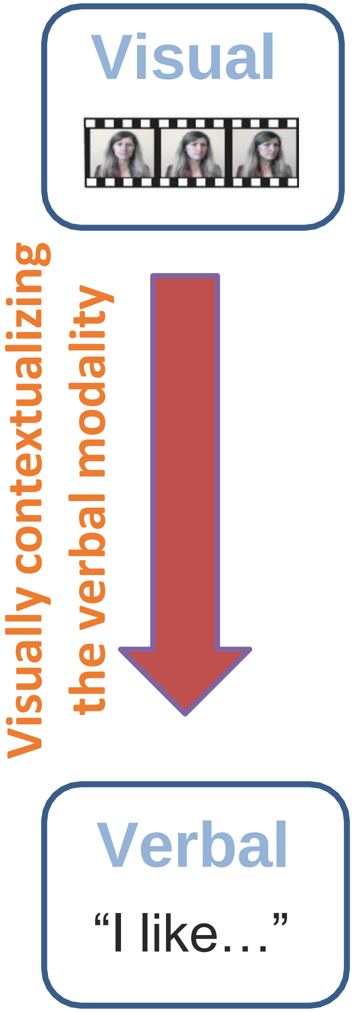


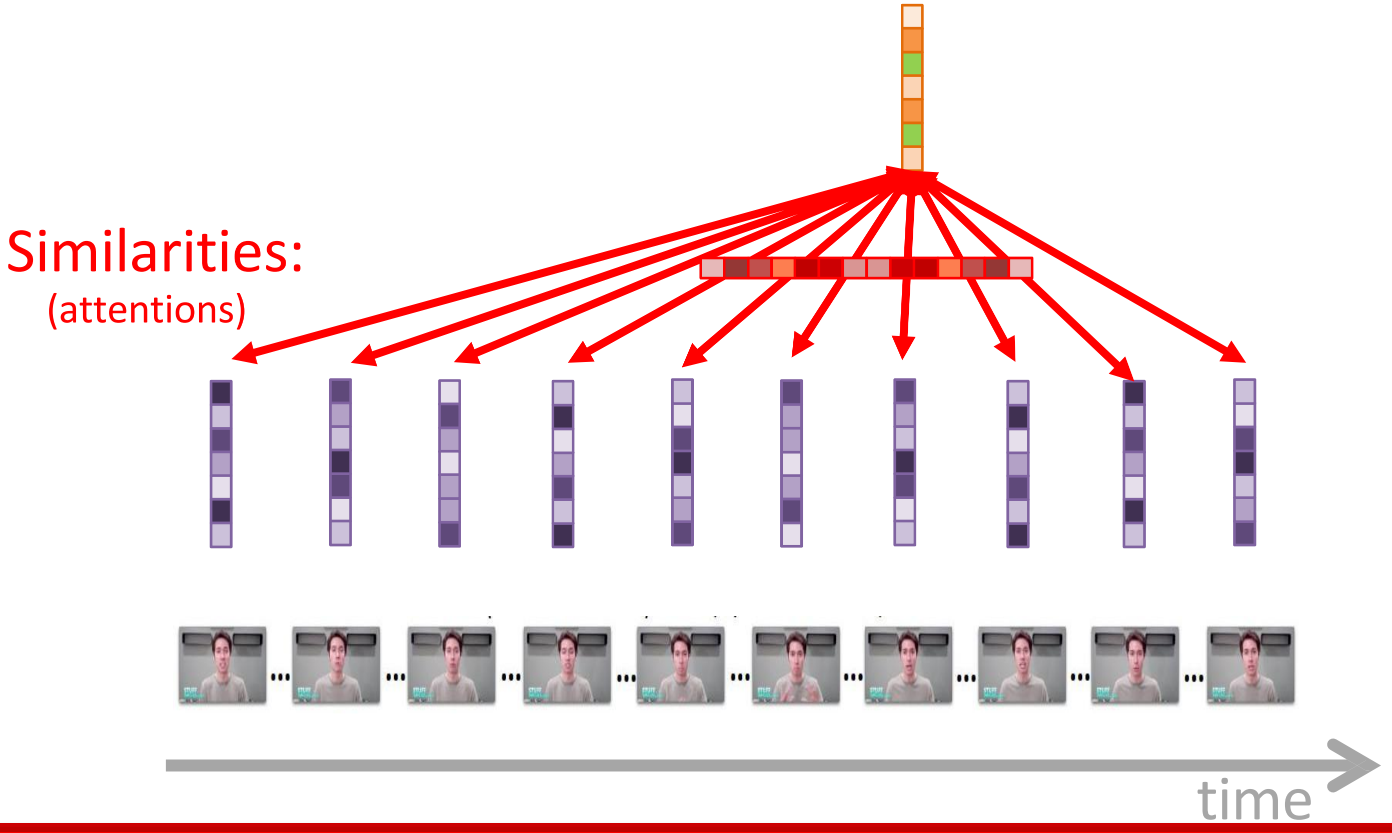
Option 2: Look at pairwise interactions between modalities

Multimodal Transformer – Pairwise Cross-Modal

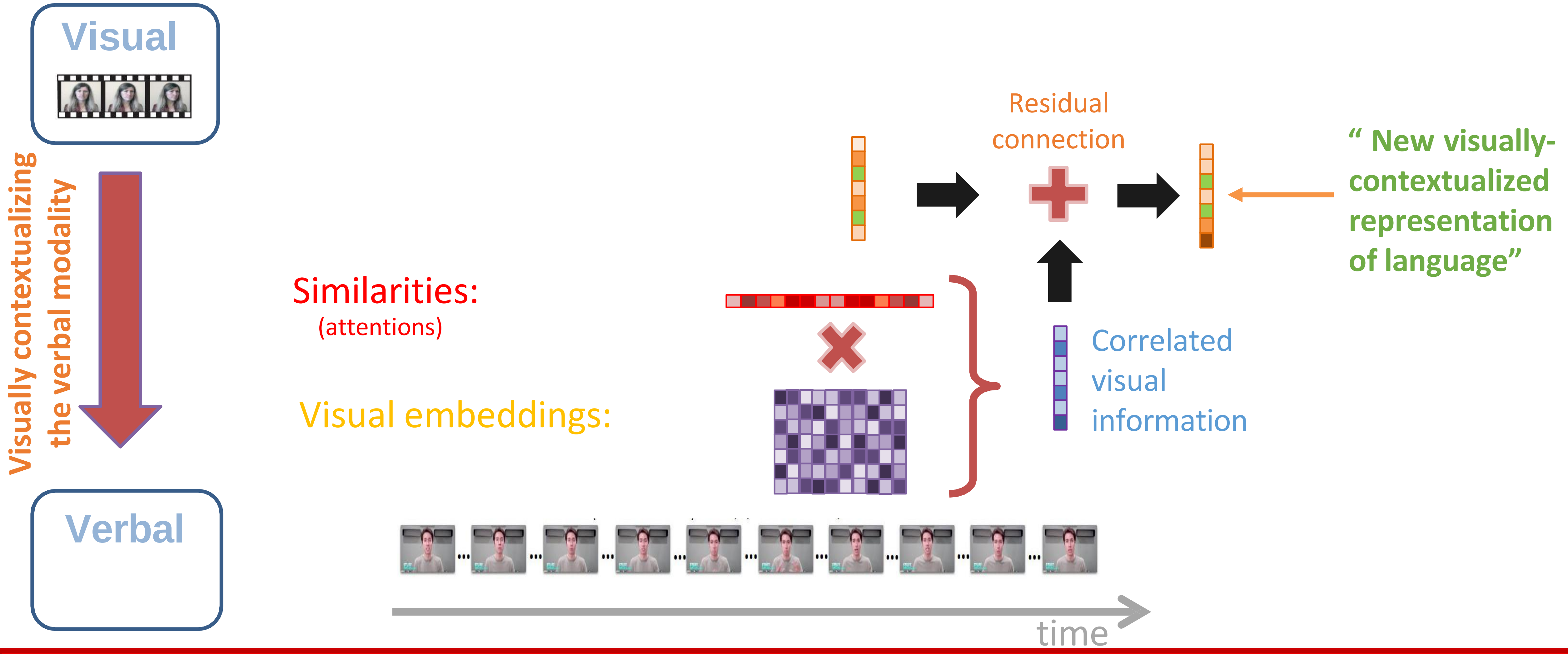


Cross-Modal Transformer Module ($V \rightarrow L$)

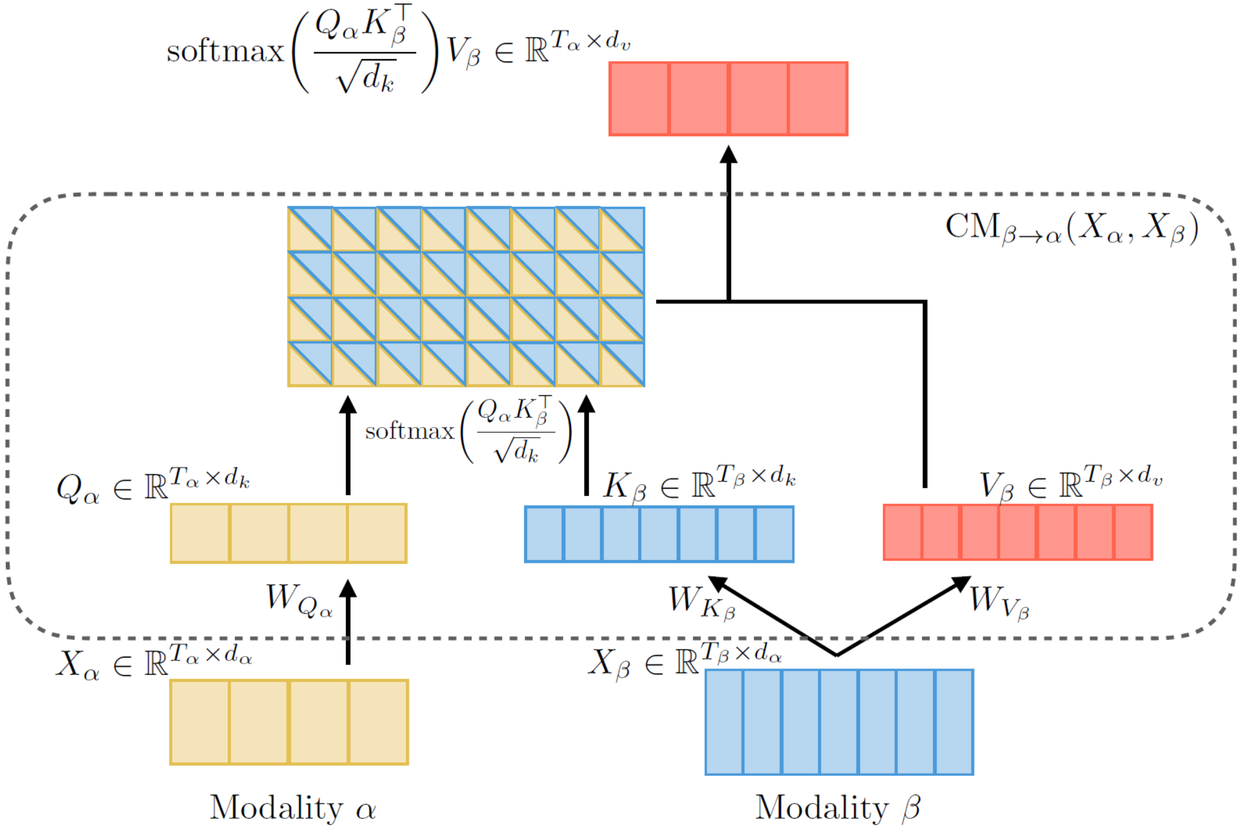




Cross-Modal Transformer Module ($V \rightarrow L$)



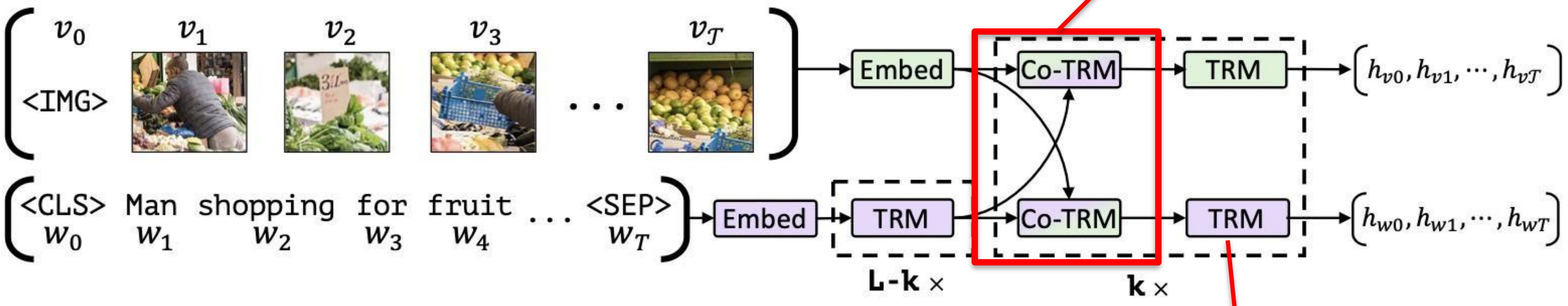
Cross-Modal Transformer Module ($\beta \rightarrow \alpha$)



Tsai et al., Multimodal Transformer for Unaligned Multimodal Language Sequences, ACL 2019

ViLBERT

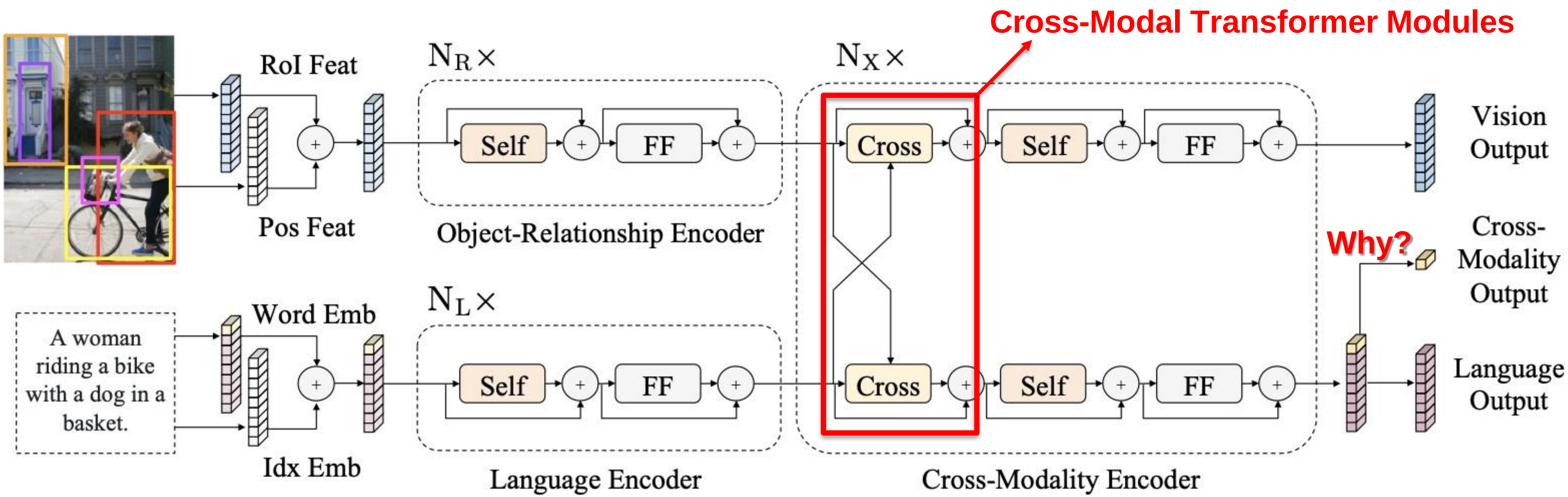
Cross-Modal Transformer Modules



Unimodal Transformer

Lu, Jiasen, et al. "Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks." *arXiv* (August 6, 2019).

LXMERT



Tan, Hao, and Mohit Bansal. "Lxmert: Learning cross-modality encoder representations from transformers." *arXiv* (August 20, 2019).

Reminder: Modality-Shifting Fusion

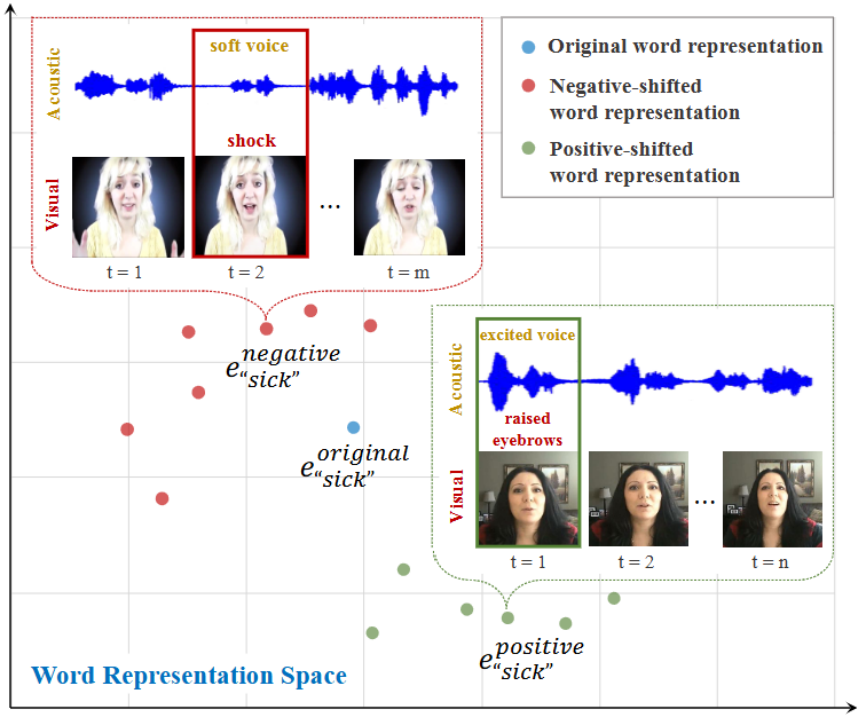


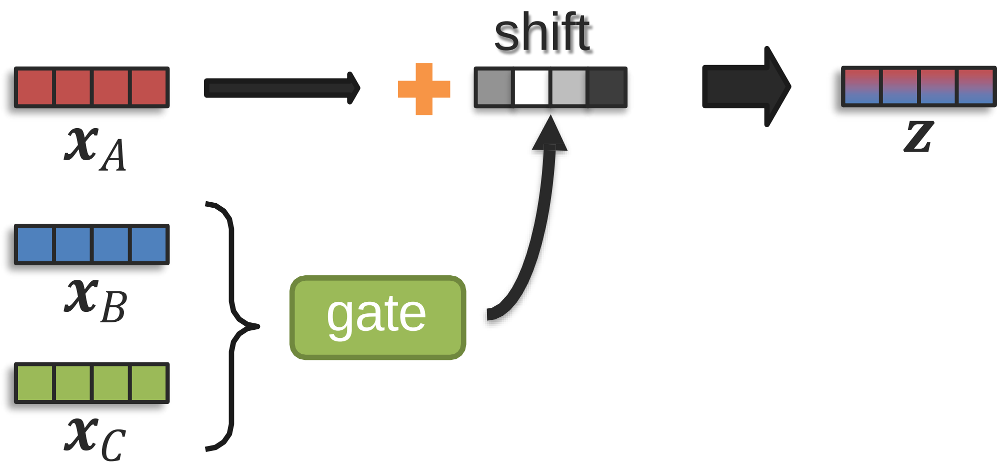
Figure 1: Conceptual figure demonstrating that the word representation of the same underlying word "sick" can vary conditioned on different co-occurring nonverbal behaviors. The nonverbal context during a word segment is depicted by the sequence of facial expressions and intonations. The speaker who has a relatively soft voice and frowning behaviors at the second time step displays negative sentiment.

Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019

Reminder: Modality-Shifting Fusion

Primary modality

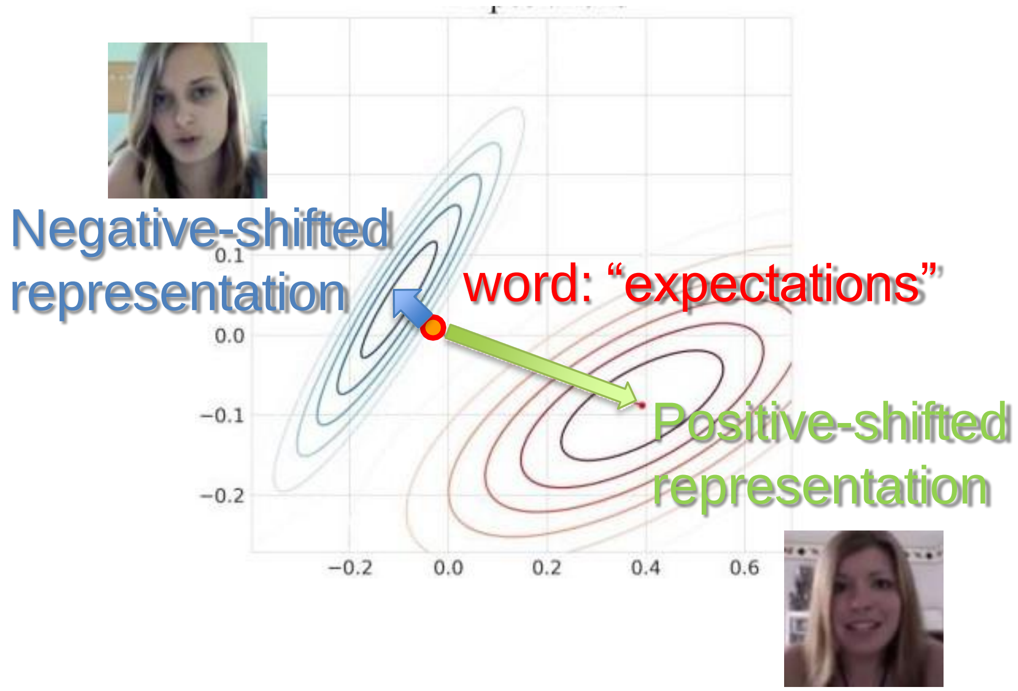
Secondary modalities



Example with language modality:

Primary modality: language

Secondary modalities: acoustic and visual



Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019

Reminder: Modality-Shifting Fusion

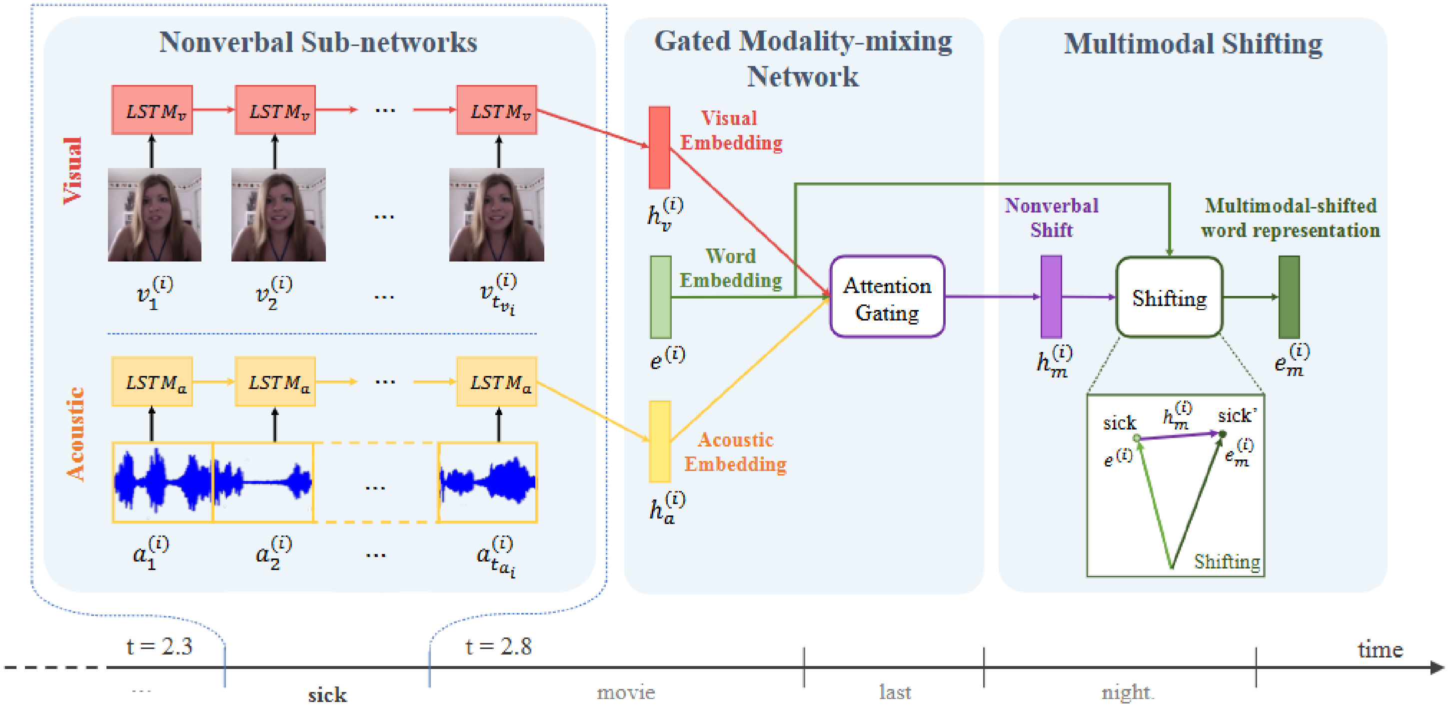
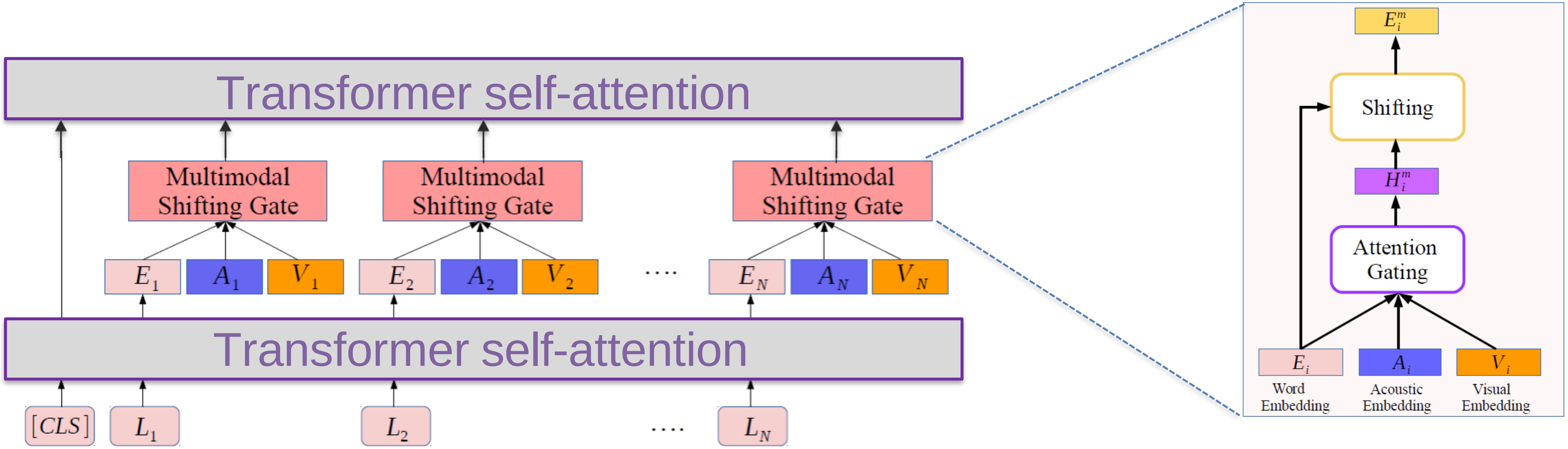


Figure 2: An illustrative example for Recurrent Attended Variation Embedding Network (RAVEN) model: The RAVEN model has three components: (1) Nonverbal Sub-networks, (2) Gated Modality-mixing Network, and (3) Multimodal Shifting. For the given word "sick" in the utterance, the Nonverbal Sub-networks first computes the visual and acoustic embedding through modeling the sequence of visual and acoustic features lying in a word-long segment with separate LSTM network. The Gated Modality-mixing Network module then infers the nonverbal shift vector as the weighted average over the visual and acoustic embedding based on the original word embedding. The Multimodal Shifting finally generates the multimodal-shifted word representation by integrating the nonverbal shift vector to the original word embedding. The multimodal-shifted word representation can be then used in the high-level hierarchy to predict sentiments or emotions expressed in the sentence.

Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019

Modality-Shifting with Transformers

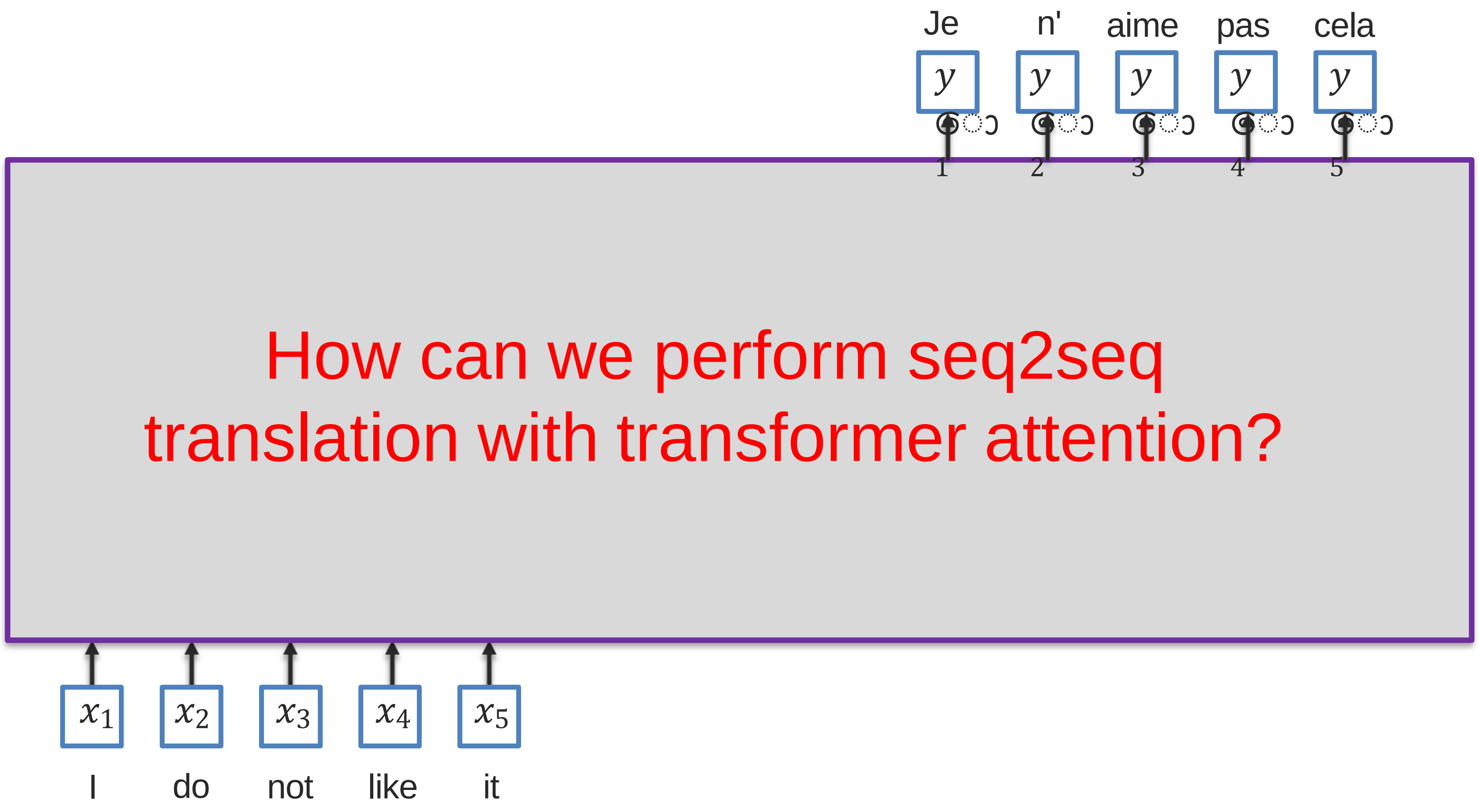
Multimodal Adaptation Gate (MAG) + BERT



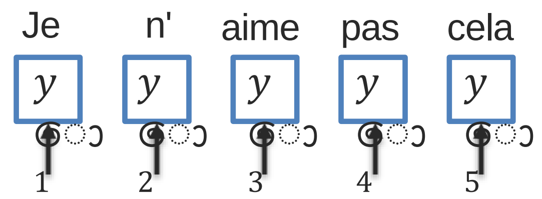
Rahman et al., Integrating Multimodal Information in Large Pretrained Transformers, ACL 2020

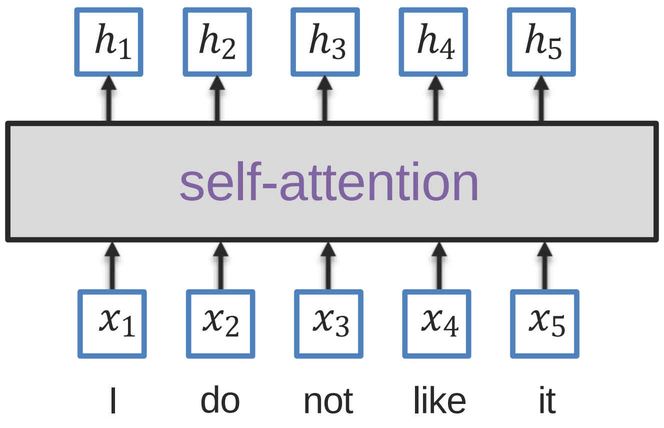
Sequence-to-Sequence Using Transformer

Sequence-to-Sequence Modeling



Seq2Seq with Transformer Attentions





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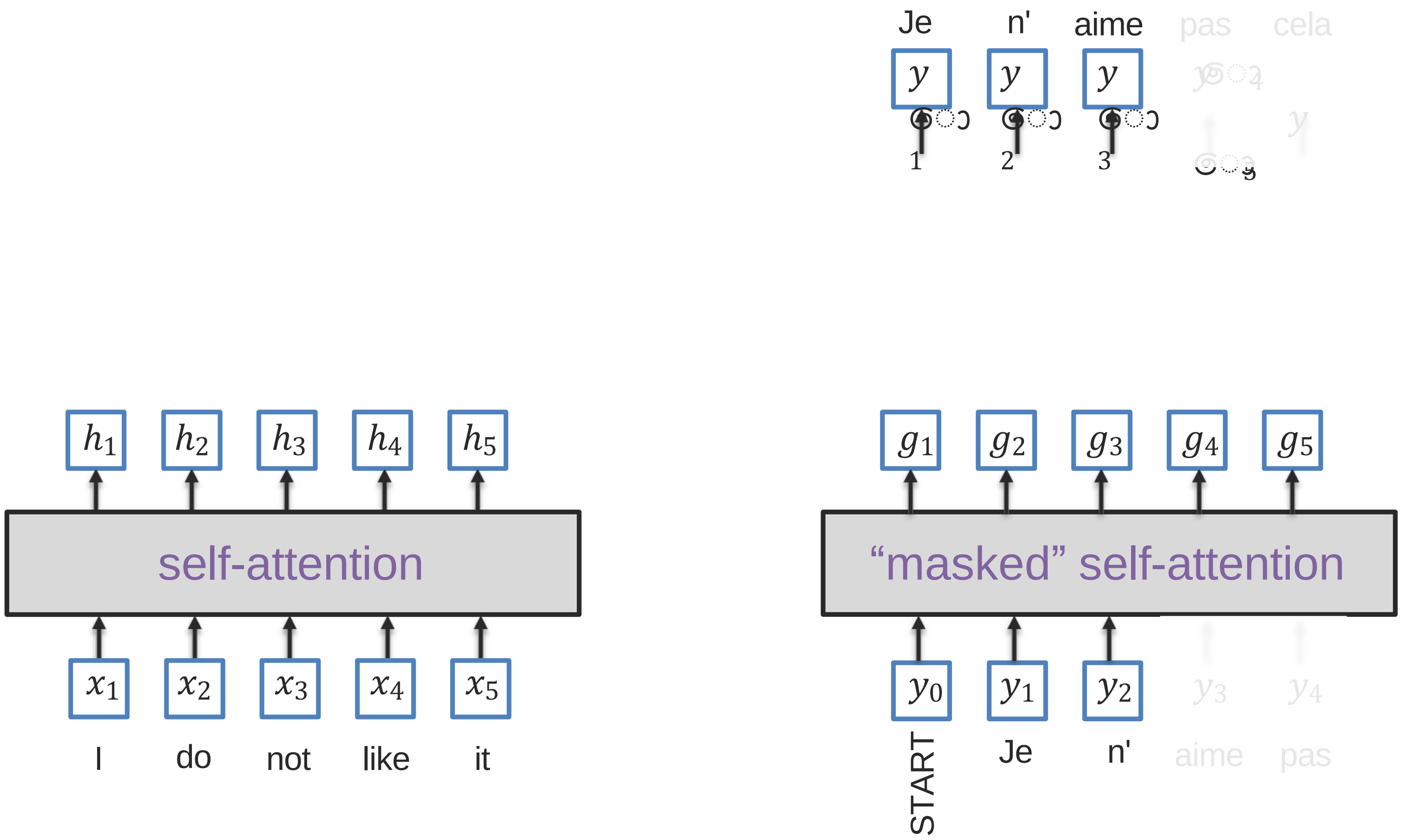
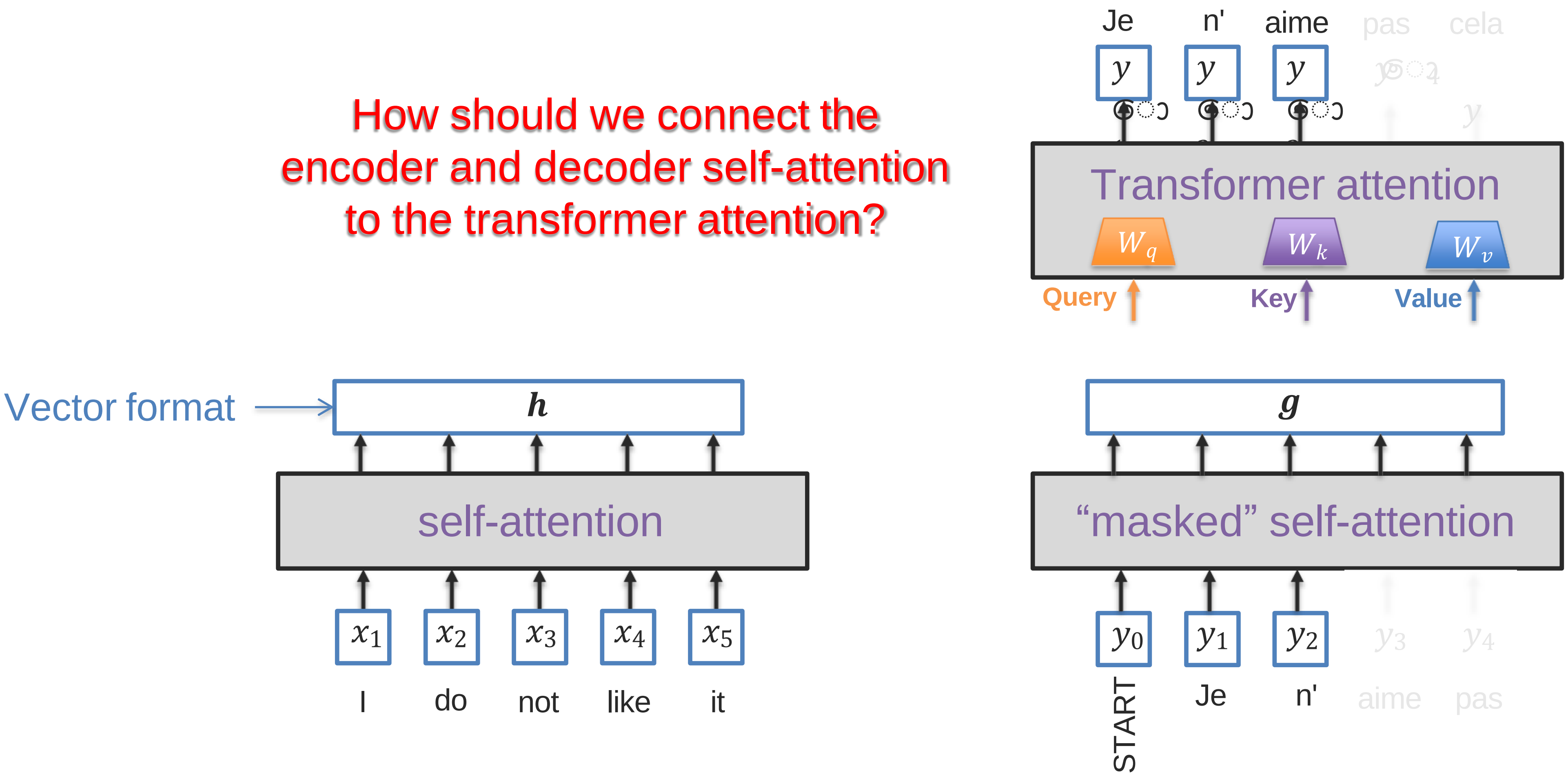


Seq2Seq with Transformer Attentions

How should we connect the encoder and decoder self-attention to the transformer attention?



Seq2Seq with Transformer Attentions